

A case study of classroom podcast in Ohlone Community College

DISSERTATION

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Abstract

This study examined students' use of podcasting and related personal entertainment technologies for learning (the frequency of classroom podcast usage) The effectiveness of podcasting was measured by students' achievement on the course midterm and final examinations. A quasi-experimental design was used. Two intact groups (classroom instruction without classroom podcasts and classroom instruction supplemented with classroom podcasts) were examined to determine the effect of podcasting on student learning. Results showed no statistically significant differences between the two groups. However, the course midterm and final examination test scores of the group with classroom instruction supplemented with classroom podcasts were slightly higher than those of the group with classroom instruction without classroom podcast, and the average final examination test scores were significantly higher than the average midterm test scores. Results also showed that the podcasting service was not for every student. The demographic profile showed that 52.5% of the students were female and 47.5% of the students were male; age groups were 19-or-younger (50.8%), 20-24 years (41.0%), 25-29 years (4.9%), and 40-or-older (3.2%). A majority of the respondents (91.8%) was in the 24-or-younger age group; ethnicity included Whites 42.6%, Asians 19.7%, Hispanics 11.5%, Multiracial, 11.5%, Filipinos 4.9%, and others 9.8%. Female students were significantly more likely to use the podcasting service. Analysis of age groups and class attendance levels showed no statistically significant differences. Although there was not much difference in the average midterm and final examination test scores for the two groups, there was evidence to suggest that students who listened to classroom podcasts were more likely to achieve higher test scores than students who did not listen to classroom podcasts. A large majority of the students had very positive views of classroom podcasts, particularly female, part-time, nontraditional, and minority students.

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A CASE STUDY OF CLASSROOM PODCAST
IN OHLONE COMMUNITY COLLEGE

A
Dissertation
Presented to the
Faculty of the Hufstedler School of Education
Alliant International University

In Partial Fulfillment
of the Requirements for the Degree of
Doctor of Education

By
Jim Pham
San Francisco 2010

Abstract of Dissertation

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IN OHLONE COMMUNITY COLLEGE

By

Jim Pham, Ed.D.

Alliant International University

Committee Chairperson: George Stamos, Ed.D.

This study examined students' use of podcasting and related personal entertainment technologies for learning (the frequency of classroom podcast usage). The effectiveness of podcasting was measured by students' achievement on the course midterm and final examinations. A quasi-experimental design was used. Two intact groups (classroom instruction without classroom podcasts and classroom instruction supplemented with classroom podcasts) were examined to determine the effect of podcasting on student learning. Results showed no statistically significant differences between the two groups. However, the course midterm and final examination scores of the group with classroom instruction supplemented with classroom podcasts were slightly higher than those of the group with classroom instruction without classroom podcast, and the average final examination test scores were significantly higher than the average midterm test scores. Results also showed that the podcasting service was not for every student. Of the 72 participants, 85% ($n = 61$) responded to the in-class handout survey. The demographic profile showed that 52.5% of the students were female; age groups were 19-or-younger (50.8%), 20-24 years (41.0%), 25-29 years

(4.9%), and 40-or-older (3.2%). A majority of the respondents (91.8%) was in the 24-or-younger age group; ethnicity included Whites 42.6%, Asians 19.7%, Hispanics 11.5%, Multiracial, 11.5%, Filipinos 4.9%, and others 9.8%. Female students were significantly more likely to use the podcasting service. Analysis of age groups and class attendance levels showed no statistically significant differences. Although there was not much difference in the average the average midterm and final examination test scores for the two groups, there was evidence to suggest that students who listened to classroom podcasts were more likely to achieve higher test scores than students who did not listen to classroom podcasts. A large majority of the students had very positive views of classroom podcasts, particularly female, part-time, nontraditional, and minority students.

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Chapter 1

BACKGROUND AND RATIONALE

The emergence of mobile technologies through the use of podcasting has significantly changed teaching and learning methods in higher education. In recent years, advancements in mobile and portable technologies, including cell phones, personal digital assistants (PDAs), and related media players (MP3s/MP4s) have revolutionized the way educators and students communicate, interact, and engage in classroom learning activities (Jham, Duraes, Strassler, & Sensi, 2008; Livoti, 2008; Reid-Martinez, Groom, & Bocarnea, 2008, p. 703). Today's podcast hosting service can be found virtually anywhere on the Internet and available on any type of mobile or portable media device. Mainstream news media outlets such as CNN, CBS, FoxNews, and NBC distribute audio/video podcasts in the Really Simple Syndication (RSS) feed format as an alternative to newspaper and Web delivery methods (Gagliolo & Nansen, 2007). Higher education institutions such as Duke University, Massachusetts Institute of Technology (MIT; 2007), and many other American colleges and universities have reported that they distribute audio/video podcasts of their professional and educational training content (Sorrentino, 2008b).

The increasing popularity of mobile and portable devices in American society has fueled the growth of podcasting in educational settings. Podcasting makes it easy for students to access, download, and playback audio/video podcasts of course content on their mobile device, PDA, or related media device (Caudill, 2007; Gillard &

Bailey, 2007; Lamb & Johnson, 2007; Stiler, 2007). Several studies have been conducted to attempt to determine the effects of mobile technologies on society. The Pew Internet and American Life Project (2008) found that 96% of Americans 18 to 29 years old reported owning a cell phone or PDA, compared to 86% 30 to 49 years old, 64% 50 to 64 years old, and 36% 65 and over. Gartner (2008) reported that people who used mobile devices were more efficient and productive than those without such devices. Gartner predicted that, by 2015, people will customize 90% of the information, tools, and educational and technological resources that they use at work, at home, and for entertainment.

This research evidence makes it clear that mobile technologies can be used as educational tools to reach out to today's generation of students. However, the benefits are not without risk. According to Carnegie Mellon (2007), the risks associated with mobile technologies are that some students may not like supplementary lesson activities through podcasts and may find instruction less interesting and meaningful. Sorrentino (2008b) also pointed out similar risks about "information fetishism" that "the abundance of digital information and communication as the exclusive means to realize knowledge, substituting them to the variegated direct experience of the world" (p. 218). It is presumed that mobile learning is analogous to formal learning but there is little evidence to predict the differential in academic performance by students. Nevertheless, people living in the 21st century cannot risk computer illiteracy in the world of the digital economy (Frabboni, 2008, p. 284).

The pervasiveness of mobile technologies (and previously the Internet and Web technologies) has brought about rapid changes in higher education. Teachers and students are embarking on a new era of college education in which technologically and socially rich environments are affecting the way educators and learners communicate,

engage, exchange information, and interact. This new era offers flexible learning opportunities for students and empowers students to take responsibility for their own learning.

According to Baird and Fisher (2005), social networks such as Facebook, MySpace, Twitter, and blogs are changing the way people communicate and relate to others. For example, during the 2009 Iran presidential elections, the world witnessed the power of these social networking sites for distributing audio/video media content and reporting news and information as it happened, when all major news media outlets were prohibited in the country. Facebook, Twitter, YouTube, and bloggers were the main sources of information and the eyes for the world; without these social networking sites, the protests against the presidential election would not have been known to the world. These communication methods provide unlimited amounts of information to people around the world, serving as the fastest ways to create, distribute, and consume information. Like the social networking community, podcasting utilizes the Internet to provide easy access to create, package, and distribute media content as audio/video podcasts.

These social networking sites and technological advances in mobile technologies have rapidly changed the face of higher education. The popular media players such as Apple® iPod®, Apple iPhone™, Microsoft® Zune®, and many other brands of MP3/MP4 (MPEG-1 Audio Layer 3 and 4) players offer podcasting capabilities to communicate, engage, exchange, interact, and distribute digital media content to the social networking community. These podcasting capabilities enable teachers and students to engage in classroom instruction in more meaningful ways. The podcasting learning experience creates flexible learning opportunities for students and promotes knowledge sharing and collaboration between teachers and students.

According to the Pew Internet and American Life Project (2005), almost 9 in 10 teens were Internet users and approximately 11 million teens went online daily. About 87% of teens in the United States used the Internet—about 21 million youth, up from about 17 million in late 2000. The numbers of teenage use of the Internet and related technology have increased in a variety of ways, including higher education. A report by the Pew Internet and American Life Project (2009b) showed that mobile telephone use had climbed 63% in the 12- to 17-year-old group from fall 2006 to 71% in early 2008. They also reported that 74% of teens owned an iPod or MP3 player and 60% of teens owned a personal computer.

Mobile, Internet, and Web technologies have seen significant growth in the past decade, with rapid growth in use by the teenage population. Ohlone College, like many other higher education institutions, has undergone a major paradigm shift from traditional classrooms to technology-mediated teaching and learning (Ohlone College, 2006; Vanin, Castelli, Pepe, & Addimando, 2008, p. 1). A direction in higher education has appeared to shift toward a blended approach of education and technology (Ng, 2008, p. 21; Ololube, 2008, p. 100). The Internet and related Web technologies have expanded educational offerings from classroom instruction to Web-based courses through the use of mobile and portable devices.

Traditionally, American higher education has adapted societal changes and technological changes, sometimes without fully understanding the effects of mobile and portable technologies and their application in education. Such lack of understanding could have unforeseen consequences for traditional forms of teaching and learning. Using podcasting to reinforce students' learning is important, but the more important issue for American community colleges is to provide the necessary resources to improve overall student performance. Current trends in mobile

technologies are just one of the ways to reach out to students. According to Ben Bernanke, Federal Reserve Bank Chairman, “The primary vehicle for decreasing income inequality and increasing the productivity of America’s workers is to provide greater access and success in postsecondary education” (as cited in Education Commission of the States, 2008, p. 1). Bernanke predicted that American community colleges will play a central role in the future of the American work force.

Statement of the Problem

Podcasting is a relatively evolving technology that describes the manner in which audio and/or video content may be accessed from cell phones, portable media players, and personal computers. According to Brown and Green (2007-2008) and Podcasting News (2008), podcasting is considered to be one of the fastest-growing technologies and it appears to be growing exponentially. Recent advances in podcasting presentations have expanded from personal entertainment usage to educational purposes.

With a rapid growth in the general population and higher education, podcasting has fundamentally changed the classroom teaching and learning environment. Many academic institutions, including Duke University, Harvard University, MIT, and Stanford University, have already offered course materials via audio/video podcasts (Chan, Lee, & McLoughlin, 2006). *The Chronicle of Higher Education* (Read, 2005) reported that in 2004 Duke University distributed iPods to all first-year students as part of an initiative to encourage the use of podcasting in education. After evaluating the program, the university revolutionized the adoption of the iPods or simply MP3 media players as a course content dissemination tool. Not only is podcasting being offered at leading universities; Ohlone College, the site for this study, launched its

own Ohlone's iTunesU, aimed to provide the capability for students to access and download audio and video podcasts to their computers or iPods directly from iTunes (Ohlone College, 2006).

Ohlone College is a part of the Ohlone Community College District, with campuses in Fremont and Newark, in the eastern part of the San Francisco Bay area in northern California. The district encompasses large cities such as San Francisco, San Jose, and Oakland. To the south of Fremont is the Silicon Valley, best known for the silicon chip innovators and the headquarters of such leading high-technology manufacturers as Apple Computer, Ebay, Google, Hewlett Packard, Intel, Oracle, Sun, Yahoo, and many more. The geographical location of Ohlone College makes a good site for this study.

According to Ohlone College's Institutional Research Office (Ohlone College, 2009), the student enrollment for the fall 2008 semester was 12,842, an increase of 9.1% from fall 2007. Ohlone offers 180 academic programs and employs about 465 part-time and full-time faculty and 235 support and management personnel. Every year more than 500 students transfer to 4-year colleges and universities and about 600 students graduate with degrees or earn vocational certificates. The demographics of Ohlone Community College District students include 3.5% African American, 35.3% Asian, 8.8% Filipino, 19.4% Hispanic, 1% Pacific Islander, 29.2% White, 2.8% other ethnic groups.

Ohlone College, like most American community colleges, is geared toward nontraditional students (adult learners and lifelong learners), rather than traditional college students in 4-year colleges and universities. According to a report on student demographics by the National Commission on Community Colleges (2008), the average age of students at Ohlone was 29; 43% of the students were in the 21-or-

younger age group and 42% were in the 22-39 age group; the 40-or-older age group was 16%; 59% of the students were women, 34% were minorities, and 8% were non-U.S. citizens. Traditional college students are typically in the 18-24 age group and nontraditional students are in the 25-and-over age group. Furthermore, community colleges serve students who are academically underprepared, economically disadvantaged, the first in their family to attend postsecondary education, or members of traditionally low-performing subgroups of students. There is a digital gap between traditional and nontraditional students.

As reported by the National Commission on Community Colleges (2008), The Education Commission of the States (2008), and Ohlone College (2009), the data suggest growing numbers of nontraditional students enrolled in American community colleges. The data showed that traditional college students also enrolled in community colleges in large numbers. The National Center for Education Statistics (2007) projected a rise of 15% in enrollments of people under 25 and a rise of 21% in enrollments of people 25 and over by 2016. These data suggest that nontraditional students are returning to school at a rapid rate. Ohlone College, like most of the California community colleges, serves students affected by the digital divide. Podcasting is just one of the ways to overcome the digital divide between traditional and nontraditional students, especially economically disadvantaged students.

Podcasting is simply the process of packaging/distributing training course contents into audio/video podcasts. Podcasting allows faculty to create, publish, and post course content on Web sites (Barlow, 2008; Mikat, Martinez, & Jamesa, 2007). Students can access and download classroom podcasts either from faculty Web sites or from Ohlone's iTunesU, to play back on their cell phones, iPods, iPhones, MP3/MP4 players, or personal computers. The additional time to review classroom lectures,

lessons, and assignments afforded by classroom podcasts offers students an alternative reinforcement for learning key concepts for the course. Based on the theories of humanism and behaviorism, reinforcement is imperative to student learning outcomes by helping students to achieve competency in what is being learned (Salkind & Rasmussen, 2008, p. 15). This theoretical framework is an effective way for educators to communicate curriculum and other vital information to students without the constraint of the classroom (Kui & Mengmeng, 2007). According to Barlow (2008) and Fryer (2008), students are likely to be more engaged in subject matter when they feel more interesting in the learning content with which they are familiar.

Not everyone agrees that podcasting is an effective tool for the classroom. Some educators maintain that podcast recording of lecture materials may not be effective in facilitating classroom learning (Copley, 2007; Farivar, 2007). Copley conducted a pilot study on the production of audio and video podcasts of lectures for campus-based students. This pilot study indicated that podcast recordings of lectures were not effective in facilitating mobile learning. As Frydenberg (2006) pointed out, teacher/student relationships and classroom attendance may have been impacted by the broad availability of podcasts because students were able to download classroom materials without having to attend classes.

Although learning through podcasting methods offers new flexible learning outside the classroom, it also presents challenges to traditional on-ground classroom learning. One of the major challenges facing classroom learning is how to engage the diverse learning styles of student, especially community college students (Lyles, Robertson, Mangino, & Cox, 2007). With the flexibility of classroom podcast learning, students choose their preferred learning styles to succeed in their overall

course work. However, the benefits of incorporating classroom podcasts into traditional classroom settings remain unclear.

In an effort to address these concerns and the effectiveness of podcasting in the classroom teaching and learning settings in Ohlone College, this study aimed at investigating whether the use of podcasting resulted in higher test scores.

Purpose of the Study

The purpose of this study was to investigate the effect of classroom podcasts as supplemental course materials for playback of lectures and other lessons in regard to the average midterm and final examination test scores of the students. The study also examined the effects of classroom podcasts and the use of mobile and portable media devices, including iPods and MP3/MP4 players, and the impact of these devices on student learning. This study is necessary to determine whether the use of classroom podcasts is consistent with traditional practices of providing meaningful learning experiences and whether there are statistically significant differences between students who receive course information through podcasts and students who do not receive course information through podcasts in relation to their test scores. In an effort to understand the effectiveness of podcasting in the classroom teaching and learning settings at Ohlone College, this study aimed at investigating the use of classroom podcasts. The aim was to determine whether the use of podcasting resulted in higher test scores. The supposition was that students who listen to classroom podcasts are more likely to achieve higher test scores than students who do not listen to classroom podcasts. It was assumed that, when students feel comfortable with mobile technologies and the use of podcasting in the teaching and learning environment, they will be more likely to adopt podcasting as a teaching/learning tool.

Research Questions

Five questions were examined to determine the effect of classroom podcasts as supplemental course materials for playback of lectures and other lessons on midterm and final examination test scores. Question 1 was the main research question for this study. Questions 2 to 5 were used to determine within-group (classroom instruction supplemented with classroom podcasts) differences.

1. Are there significant differences between two types of educational instruction (traditional classroom instruction and classroom instruction supplemented with classroom podcasts) in regard to the average midterm and final examination scores of students?
2. Is there a significant difference between the numbers of times students listen to classroom podcasts before and after the midterm?
3. Is there a significant difference between class attendance (less than other classes, same as other classes, and more than other classes) with regard to the number of hours students listen to classroom podcasts per week?
4. Is there a significant difference between age groups with regard to the number of hours students listen to classroom podcasts per week?
5. Is there a significant difference between genders with regard to the number of hours students listen to classroom podcasts per week?

Significance of the Study

This study investigated the effect of classroom podcasts in regard to the average midterm and final examination test scores of students. This study also determined the students' attitudes toward classroom podcasting in relation to classroom podcast usage. The main focus for this study was on the frequency of

classroom podcast usage. The significance of this study was whether the use of podcasting effectively improves student learning. The idea of podcasting represents new thinking in the delivery of educational content for playback and/or viewing lectures and lessons on student personal media players. Classroom podcasts have the potential to enrich existing classrooms and free students from many learning constraints. Podcasting in the learning environment reinforces what students are learning and exposes them to enhanced learning of the subject matter, thereby improving their understanding of course materials. The learning outcomes for students through the use of podcasting could enhance the traditional classroom as well as online learning avenues by providing additional learning materials.

Despite the benefits of podcasting, ultimately the learning outcome depends on the ability of the student to retrieve course materials and supplemental instructions made available to them. First, the findings of this study could be used to help educators to determine the necessary balance between podcast learning and traditional classroom interaction. Second, the findings from this study could be used to determine whether podcasting (production, publication, and delivery of audio and video files) is an instructional path. Third, data gathered in this study could be used to provide leadership and support for future directions of technology-based instruction in higher education. Should classroom podcasts be used to support learning by community college students? The more important question for consideration is whether the use of technology effectively improves student learning. Fourth, the conclusions from this study could be used to expand technology-based learning and provide assistance to educators who desire to develop and share podcasts containing course materials with students. By understanding the benefits of podcasting, educators would have enough knowledge to determine whether podcasting (production, publication, and delivery of

audio and video files) is an instructional path they would like to undertake. The decision is ultimately based on individual judgment.

Methodology

A quasi-experimental design was used to determine the effect of classroom podcasts in regard to the average midterm and scores of students. Traditional classroom instruction measures (attending lectures, reading the textbook, and taking notes) were used in a controlled sample. Supplemental instruction through the use of classroom podcasts was compared with traditional classroom instruction to determine significant differences between the two groups.

For the scope of this study, the term *podcasting* represents the method of recording and distributing of course content, such as lectures, lessons, and assignments. Similarly, *podcast* refers to audio/video files of course content. The *production of a podcast* is the process of creating self-published audio/video content as MP3/MP4 files and making them available for download via faculty Web sites (Kui & Mengmeng, 2007; Livoti, 2008; Worcester & Barker, 2006).

The participants in this study were students from a lecture-based course, History of the United States (HIST-117A), with the maximum enrollment of 120 students at Ohlone College. This course met twice weekly. A podcasting of the classroom lecture for each of the meetings was published on Ohlone's iTunesU and WebCT. All students were invited to participate in the study. The student population included male and female students divided by age into groups of 19 years and younger, 20 to 24 years, 25 to 29 years, 30 to 34 years, 35 to 39 years, 40 to 49 years, and 50 years and older, in all class levels (freshmen, sophomore, junior, senior, and graduate or postgraduate) and from a variety of ethnic backgrounds. The distribution

of students by age group was the same as that used by the California Community Colleges Chancellor's Office (2009) and Ohlone College's Institutional Research Office (Ohlone College, 2009) to track student headcount.

A three-part in-class handout survey/questionnaire was used to examine the experience of extracurricular usage of classroom podcasts. The results were used to determine classroom podcast usage in relation to the learning outcomes for these community college students. The goal of the questionnaire was to investigate the effectiveness of classroom podcasts in relation to traditional classroom instruction.

The questionnaire included (a) demographics, (b) podcasting, and (c) open-ended questions. Demographic questions were used to collect general information about participants: gender, age group, class attendance, class level in college, and ethnicity. Podcasting questions were used to collect the frequency of use of classroom podcasts. Open-ended questions were used to collect additional information.

Students were asked to respond to the in-class handout survey/questionnaire about classroom podcast usage in relation to their test scores. For attitude questions, a Likert-type scale of 1-5 was used (1 = *strongly agree*, 2 = *agree*, 3 = *somewhat agree*, 4 = *disagree*, 5 = *strongly disagree*). For frequency of classroom podcast usage, respondents were directed to check all that applied. Only the submitted survey/questionnaire from the students were used in the study.

The collected data were coded and analyzed to explore midterm and final examination test scores, frequency of classroom podcast usage, and student attitudes regarding classroom podcasts.. The students' midterm and final examination test scores were compared to determine the effectiveness of classroom podcasts. Inferential statistics were used to determine the differences between and within groups

of students (traditional educational instruction and classroom instruction supplemented with classroom podcasts).

Theoretical Framework

Emerging mobile and podcasting technologies have revolutionized the way educators and students communicate, interact, and engage in classroom learning activities. The Internet has also created an information-rich teaching and learning tool that facilitates social interaction and exchange between educators and students. These technologies allow educators and students to access a wealth of knowledge-based information and assessment tools that expand the traditional classroom.

Several studies have been undertaken on the use of podcasts in the classroom (Evans, 2008; Frydenberg, 2006; Smith & Woody, 2000). Smith and Woody found that computer-assisted instruction had a positive effect on lectures, especially in taking notes and studying for examinations. They reported that technology-based approach in instruction was perceived as favorable by students and yielded significant improvements in student learning, based on both student self-report and objective outcome testing. Evans studied the effectiveness of mobile learning in the form of podcasting for teaching undergraduate students in institutions of higher learning. Results showed that undergraduate students embraced the mobility that iPod and MP3 players provide. Frydenberg studied classroom usage of podcasts and the students' (rather than instructors') use of podcasting on course lecture materials. Results in these studies showed that podcasting was a useful tool for disseminating course information and those students were likely to use podcasts when they were available to them.

Results of these studies revealed numerous advantages and disadvantages of the use of podcasts in classrooms for higher education. However, there was

compelling evidence from these studies to suggest that podcasting could be used as learning tool to engage, stimulate, and encourage students. The theoretical frameworks for this study were based on earlier studies on the use of classroom podcasts by Evans (2008) and Frydenberg (2006) and self-directed learning transformational learning. Self-directed learning, constructivism, and transformational learning theory suggest that students take ownership for planning, carrying out, and evaluating their own learning experience (Merriam & Cafarella, 1999, pp. 318-320). Based on transformative learning theory by Salkind and Rasmussen (2008), having experience helps the learner make meaning of the experience (p. 14).

Based on these frameworks and educational practices, students will take the primary initiative to download and listen to podcast-mediated learning in ways similar to distributed learning, e-learning, distance education, and online learning (Inoue, 2008, p. 15). This theoretical framework is consistent with Ohlone College's Vision and Mission statements (Ohlone College, 2004-2009) and consistent with Ohlone College's Learning College initiative and the work of Senge (1996). One of Ohlone College's goals is stated as follows: "Develop across the curriculum the Learning College Model, utilizing methods and technologies that hold the most promise for improving student course and program completion success rates." This initiative has intentionally restructured the curriculum in ways that bring students, faculty, and disciplines together in more integrated and meaningful ways (Reid-Martinez et al., 2008, p. 703). The use of podcasting to facilitate classroom learning aims to make deep cognitive, neural, and social connections. Learners learn best in their specific social and cultural environments where they actively engage in social construction of knowledge (Cartelli, 2008b, p. 768; Jimoyiannis, 2008a, p. 112; Merriam &

Caffarella, 1999, p. 69). Many educational researchers have suggested that this is a period of cultural politics of flexible learning (Ziguras, 2001).

Definition of Terms

Terms related to the subject of this study are widely used in today's society to describe the technology they represent. The following terms are operationally defined for application in this study.

Blended teaching and learning environment: The combination of instructional technology and pedagogical methods strategies that facilitate teaching and learning.

Business enterprise: An economic activity of providing goods and services undertaken by a commercial or industrial service.

Computer-mediated learning: Computer, projector, the Internet, and related computer technologies are used to deliver learning material and provide learners and instructors with opportunities for interaction.

Constructivism: Educational theory that suggests that, through processes of accommodation and assimilation, learners construct new knowledge from their experiences.

Constructivist learning: Constructivist learning has some basis in cognitive learning and is the result of the mental construction of a situation. Constructivism, in its most basic form, is piecing together new information using information already known to the student.

Constructivist self-directed learning: The process whereby new meanings are created by the learner within the context of her or his current knowledge.

Culture of technology: Information technology as a part of social and cultural environment.

Distance learning/education: Use of the Internet and Web-enhanced technology as an educational delivery system designed to provide flexible ways of learning for students and to promote interaction between learners and instructors without the constraint of a physical classroom.

Distributed learning/education: Use of the Internet and related computer technologies to provide flexible learning for students.

Educational podcasting: The process of creating and publishing audio and video podcasts for use in traditional classroom education.

Exploratory data analysis: The process of computing various descriptive statistics and graphs.

Information age: The time in which abundant information technology and knowledge are available to be accessed anywhere and anytime. Other names used to describe the information age include *information knowledge society*, *Internet society*, *digital society*, and *mobile society*.

Instructional technology: The use of technology to solve educational challenges, both in the classroom and in distance learning environments.

iPods: A family of portable media players (iPhone, iPod, etc.) developed by Apple Computer Inc. (2007) to play digital audio and video podcasts. The iPods' functionalities are like those of other MP3/MP4 players.

iTunes: Apple's digital media player application that is used to download digital audio and video podcast to computers or digital media such as cell phones, PDAs, or MP3 players.

MP3: MPEG-1 is a digital audio encoding file format used to store digital audio podcasts for playback on a portable media player or MP3 player.

MP4: MPEG-4 Part 14 is a multimedia container format used to store digital audio and video streams or podcast files. MP4 is also known by the filename extension .mp4.

Multimedia: A range of media using multiple forms, such as animation, text, graphics, audio, and video, to deliver learning material. Cell phones, PDAs, iPods, and related MP3 players are used to facilitate learning.

Ohlone's iTunesU: A podcasting service that allows educators at Ohlone College to publish lectures and related class information through audio podcast and allows students to subscribe and download podcasts automatically via iTunes.

Online learning/education: Any class that offers the entire curriculum through the Internet and Web-enhanced technology to provide distance learning.

Personal digital assistant (PDA): A handheld computer device that has the ability to access the Internet and that can be used with mobile phones, Web browsers, and/or portable media players.

Podcast: A prerecorded classroom lecture and learning materials produced as audio and video files.

Podcasting: A process of creating and distributing classroom learning materials through audio and video podcasts to the Internet, allowing users to receive files automatically.

Portable digital media device: Small electronic digital handheld device, such as a smart cell phone, PDA, or laptop computer.

Portable media player: A digital electronics music player such as cell phone, iPod, or other MP3 player, used to listen to classroom podcasts.

Really Simple Syndication (RSS): A family of Web feed formats used to publish frequently updated classroom podcasts through iTunes.

Social cognitive: Self-observation and enactive experiences through social learning, emphasizing self-efficacy in learning.

Social networks: Social networking communities connect people to allow information exchange. Some of the popular Internet communication services are Facebook, MySpace, LinkedIn, Twitter, and blogger.

Technology: A manner in which the Internet, information and communication technology (ICT), and related computer technologies are used to advance teaching and learning.

Virtual university: An organization that provides higher education on the Internet through Web-enabled technologies to deliver distance learning opportunities.

Web-based course management system: A software application that enables the management and delivery of learning resources to students at any time and at any place by accessing a browser such as Blackboard (<http://www.blackboard.com>), TopClass (<http://www.wbtsystems.com/>), or Moodle (<http://moodle.org/>).

WebCT: Technology-assisted learning system that enables the management and delivery of learning resources to students at any time and at any place by accessing the information via a browser.

World Wide Web (WWW): Internet service that uses iTunes and Internet browsers to publish and download classroom podcast.

Organization of the Dissertation

The report of this study is organized in five chapters. Chapter 1 describes the research study and its context, including the background of the problem, the statement of the problem, the purpose of the study, the research questions, the importance of the study, the methodology, definition of terms, and the organization of the study. Chapter

2 reviews the literature and related research in the areas of (a) technology and society, (b) the impact of technology on college education, (c) the impact of technology on teaching and learning, (d) the use of podcasting for teaching and learning, and (e) learning theories, practices, and earlier studies in the area of technology-mediated teaching and learning. Chapter 3 describes the research methodology and the design used in classroom podcasts for community college. Chapter 4 summarizes and discusses the findings of the study. Chapter 5 provides conclusions, recommendations for further research, and implications for practices. Appendices outline the variables used in the study and contain reports on the effective use of classroom podcasts. These chapters are based on the elements of a dissertation suggested by Cone and Foster (2006, pp. 83-92).

Chapter 2

LITERATURE REVIEW

This chapter reviews the literature in educational research on the Internet and Web technologies that have revolutionized teaching and learning in higher education. The chapter examines current information on mobile technology trends and how these technologies as well as podcasting are utilized in teaching and learning in today's information age and higher education. Five current technology trends are discussed to address the benefits of today's technological changes and lifestyles and learning styles of students: (a) technology and social network communities, (b) the impact of technology on higher education, (c) the impact of technology to impart learning, (d) the use of podcasting for instruction, and (e) learning theories, practices, and earlier studies in the area of technology-mediated teaching and learning. Mobile devices (MP3, MP4, cell phone, PDA, or related media devices), as well as podcasting, have emerged as the preeminent tool for teaching and learning in most academic institutions. Without thorough understanding of the current technology trends, it will be difficult to justify the necessary balance between podcasting and the traditional forms of classroom instruction.

This chapter shows that, by leveraging podcasting in the classroom setting, students are motivated, engaged in the learning process, and more apt to perform better and participate effectively in collaborative coursework. The philosophical approach to education through the use of podcasting is to distribute classroom

materials to students via personal computers, Ohlone's iTunesU (Apple's iTunes software application), and MP3 devices. It is not the intention of this study to change or replace traditional forms of teaching and learning, but rather to encourage podcasting as a teaching alternative to relay classroom instructions or as a supplemental instruction to disadvantaged students. The approaches utilize mobile technologies and foment outreach by adapting to technological culture of the student population.

Current trends in mobile and podcasting technologies have gained attention because of the capability to distribute audio/video on demand for playback. Mobile and podcasting technologies are relatively new movements in today's higher education forms and have already been woven into today's social fabric (Kim, Mims, & Holmes, 2006). Mobile technologies receive wider adoption from many of the mainstream news media outlets, social network communities, bloggers, and information exchanges. These formats are now becoming the foremost technologies in the education market and are particularly very popular among young people, including college students (Fini, 2008, p. 714). Altbach, Berdahl, and Gumport (2005) argued that "students consumerism is a central part of the ethos of American higher education" (p. 307). Altbach noted that education in the United States has focused primarily on students and the lives of students. Based on these assumptions, American higher education institutions generally will be more likely to embrace students' culture to bring about reform in education. Podcasting is that new technological culture.

Pedagogical approaches based on these assumptions are essentially creating new alternative learning for gathering, customizing, and sharing information among teachers and students such that they reach out to other students in disparate localities (Kidd, 2008, p. 847; Vequist, 2008, pp. 87-88). As result of mobile and podcasting

technologies, the academic environment has shifted from traditional classroom settings to mobile learning centers. A study by the Pew Internet and American Life Project (2008) found that 96% of Americans 18 to 29 years old reported owning a cell phone or PDA. Innovation in mobile technologies has fueled the ability to distribute audio and video podcasts to play back on mobile devices (Sorrentino, 2008c, p. 512).

The art of mobile centric and podcasting technologies has emerged in the form of new education tools gaining widespread acceptance in higher education (Kim et al., 2006). Podcasting allows students and educators to communicate, interact, and engage in classroom instruction and collaboration in more meaningful ways (Jham et al., 2008). This pedagogical practice is consistent with traditional forms of classroom education where students attend lectures, take notes, and review classroom reference materials on their cell phones, PDAs, MP3s, or MP4 players. Instead of focusing on the lecturers while taking notes in the classroom setting, students can schedule to attend in person or engage in lecture discussions rather than focus on note taking. This method allows students and teachers to banter in a more interactive and collaborative learning environment and still be able to review classroom podcasts subscribed on their mobile devices or any of the related media player devices. The power of mobile centric and podcasting technologies, combined with pervasive access to the Internet and Web technologies, delivers distance education in American colleges and universities, European schools, and advanced academic institutions around the world, thereby creating new avenues to deliver flexible and boundless learning in today's culture of abundant streaming technologies and captive instructional styles (Inoue, 2008, p. 17; Ng, 2008, p. 21; Ololube, 2008, p. 100).

Technology and Society

Technology advances of the 1990s and the broad accessibility of the Internet have profoundly changed and transformed traditional society into an information society. These technological advances and innovations have transformed every aspect of society, especially in business, higher education, government, and the global industrial complex (Sorrentino, 2008a, p. 32). Some widely used terms to describe the abundance of information and knowledge database accessible in the information society are *subject matter knowledge society*, *Internet society*, *digital society*, and now *mobile-centered society*. Information and communication technologies remain the key factors in any organizational development, particularly in higher education. While incorporating the state of art technologies in their business, organizations can function more effectively and efficiently and can outperform competitors (Chen, 2008, p. 401). Academic institutions are not-for-profit businesses; however, having these technologies means that they will be able to provide better tools for communication and collaboration, in addition to teaching and learning. Many higher educational institutions worldwide have already adopted the advanced technologies to provide better communication and collaboration between stakeholders and teaching and learning (Inoue, 2008, p. 14).

During the past two decades enhanced communications technologies, such as faster computing power and advanced Internet capabilities, have become the driving force in allowing people worldwide to communicate and exchange information quickly (Cartelli & Palma, 2008, p. xiii). Businesses, government, higher education, and global industries have utilized information and communication technology to streamline operations and function more effectively and become efficient (Ololube, 2008, p. 100). Advances in computer technology, computer networks, computer

software and applications, and the Internet have created a means of communication for these organizations to interact with their stakeholders more effectively and efficiently. With the Internet as a centralized network, people throughout the world can send and receive information to anyone and that information can be used at home, at work, and around the world.

Today these computer services are widely available on personal computers, handheld devices (PDAs), and mobile devices that make it possible for individuals and organizations to interact and exchange information. These technologies are widely used to enhance students' learning experiences in higher education. Similarly, teachers, learners, and parents can participate and interact together in a learning community (Hussain, 2008, p. 72). The daily lives and lifestyles of every person in the information society have been greatly improved by the advent of personal computer, the Internet, and most recently, mobile technology. Increasingly, mobile devices have become the common means for people to exchange information from any geographic location in the world. Emerging mobile technologies are considered to be primary and their advances will change the way people live (Kunz & Gaddah, 2008, p. 25).

According to the Pew Internet and American Life Project (2009a), Internet and mobile phone users tend to have larger and more diverse networks of close confidantes than do those who do not use the Internet or mobile phone. They found that the use of social networking services such as Facebook and Twitter in particular were more associated with more diverse social networks. The reality of the information society is that new communication technologies allow and even encourage people to interact with other people. As social networking and Web technology enabled a new way of communicating, structuring, and sharing knowledge, podcasting interacts between learners and knowledge. Multimedia (audio, video, graphics, and text) can be

combined to deliver an interactive learning environment that encourages “individuals to learn together with others, maintaining their control over time, space, presence, activities, identity and relationships” (Anderson, 2005, as cited in Fini, 2008, p. 713). Podcasting is that new multimedia-based instruction.

Similar to the information society, advanced technologies are also transforming the global economy; emerging mobile technologies have brought significant changes in the information age to the global society transformation. In recent years, mobile devices have become much more technically sophisticated. Mobile devices such as iPhones, smart phones, and PDAs have integrated wireless capabilities that allow students to access the Internet and perform various network-related functions (Kidd, 2008, pp. 848-850). Increasingly, these devices are becoming multipurpose devices for daily routine tasks, especially for personal, business, and education activities (Kunz & Gaddah, 2008, p. 25). The daily lives of everyone are increasingly dependent on mobile devices for communication, entertainment, information, and managing everyday needs (schedules, contacts, etc.). Mobile technologies have brought about rapid changes in today’s higher education and the larger society. Gartner (2008) found that people who used mobile devices were more efficient and productive than those who did not use such devices. They predicted that, by 2012, one quarter of the world population would be using mobile devices. All signs point to mobile computing having an even greater footprint in the future of higher education while becoming more entrenched in everyday life (Sorrentino, 2008a, p. 36). This is essentially what Francis Parker (1837-1902), a father of U.S. progressivism, implied: Learners should have direct experience with objects in the environment. Mobile computing and other technological advances have transformed the art of teaching and learning (Gutek, 1997, p. 70).

To prepare for the future information society and in collaboration with International Telecommunications Union (ITC), the United Nations (UN) organized the first World Summit on the Information Society (WSIS) to provide guidance for applying information and communication technologies to benefit humanity. The first principle of the Geneva declaration stated:

We, the representatives of the peoples of the world, assembled in Geneva from 10-12 December 2003 for the first phase of the World Summit on the Information Society, declare our common desire and commitment to build a people-centred, inclusive and development-oriented Information Society, where everyone can create, access, utilize and share information and knowledge, enabling individuals, communities and peoples to achieve their full potential in promoting their sustainable development and improving their quality of life, premised on the purposes and principles of the Charter of the United Nations and respecting fully and upholding the Universal Declaration of Human Rights. (WSIS, 2003)

The Impact of Technology on Higher Education

Since the 1990s Internet and Web technologies have revolutionized world commerce, communication. The WWW has reached a critical mass that touches the lives of every individual and business. These technologies have also greatly altered the academic landscape of higher education (Ololube, 2008, p. 100). Internet and Web technologies have created opportunities for exchange information, technology-facilitated teaching and learning, as well as distance learning for higher education. Based on the ideas of Readings (1996, as cited in Robins & Webster, 2002), Robins and Webster argued that technological changes have transformed the “university of excellent, alias the technological university or the corporate university” (p. 4).

Today’s higher education institutions worldwide offer academic programs in various distinctive forms: (a) classroom education, (b) distance education, (c) online education, and (d) hybrid education (Cartelli, 2008a, p. 350). Like the Internet and

Web technologies, emerging mobile technology adds new dimensions for students to learn, communicate, and exchange information. A study by the Pew Internet and American Life Project (2008) found that large percentages of college age adults (86% of those ages 30 to 49 years, 64% ages 50 to 64 years, and 36% ages 65 and over) reported owning a cell phone or PDA.

College students are increasingly dependent on mobile technologies for learning, communication, entertainment, and exchange information. As Internet and Web technologies have helped businesses and industries worldwide to operate more effectively and efficiently, mobile technologies are similarly having a revolutionary impact on the educational system (Chen, 2008, p 401; Sorrentino, 2008a, p. 39). Research has shown that American universities, European schools, and academic institutions around the world are increasingly adopting the same kind of technologies as in business enterprises for teaching and learning, in addition to operational efficiencies, communication, and information dissemination (Raisinghani, 2008, pp. 526-528). Classroom instruction environments have also blended the use of these technologies to facilitate, publish, and distribute learning materials. Some of the commonly used technologies for classroom education include computers, projectors, presentation utilities, the Internet, faculty Web sites, email, and Web-based course tools (Lee, Pillay, & Chandra, 2008, pp. 532-535).

A study by the University Centre for Research, Teaching Innovation and Distance Learning (CARID) found that technology-enabled learning was coming closer to and interacting more with knowledge databases (Poletti, 2008, p. 96). Findings suggested that technology learning focused on each individual person to manage his/her learning needs. A study by UCLA found that personal knowledge management concentrated on five main tasks: (a) searching/finding, (b) categorizing/

classifying, (c) naming things/making distinctions, (d) evaluating/assessing, and (e) integrating/relating (Sorrentino, 2008c, p. 512). These two principles suggest that individual students or educators are responsible for planning and mapping out their own teaching/learning experiences. These principles are consistent with self-directed learning, constructivism, and transformational theories that suggest that individuals take ownership for planning, carrying out, and evaluating their own learning experiences (Merriam & Cafarella, 1999, pp. 318-320). Gallerani (2008) also aligned his theory of self-constructed autonomy (p. 520). The self-constructed autonomy defines the meaning of the free explanation of the training path of every individual and the constructed dependence defines the other side, the indissoluble/relentless rooting in a net of relations inside a community. This also suggests that learning is considered as both social and cultural aspects of society.

These technology adaptations have been accepted as indispensable parts of industrialized societies. They must further adapt to meet the challenges of the knowledge age (Corazza, 2008, pp. 357-360). American higher education has been traditionally responding to changing student curricular interests by expanding fields and departments and cutting unpopular areas; thus, students' interests can affect academic policy and governance (Altbach et al., 2005, p. 307). Adapting to technological change is a natural process of human behavior.

The future technology trends are expected to grow even more rapidly than the Internet and will continue to shape the nature of the workplace and higher education. Many scholars in the area of educational research believe that future trends of the education system in the 21st century are linked to technological advances (Inoue, 2008, p. 17; Vanin et al., 2008, p. 3). As Poletti (2008) pointed out, technology is "a new reality" in higher education where methodological practice can be used in a

variety of public and private sectors, ranging from school and university to professional and corporate training. Poletti suggested that the world of education and student learning training is becoming a new way of structuring, sharing knowledge, and the culture from which such knowledge derives and that it must be offered to the society of knowledge (pp. 95-96). This suggests that technology is a structured learning process that encourages a sharing culture and the acquisition of new knowledge.

A 2003 study by Boettcher (as cited in Lee et al., 2008) focused on online learning systems. Boettcher argued that the structure of the learning content is vital to its success. She listed four key core components of the learning experience: (a) the course material to be learned, (b) teaching, (c) the learner, and (d) the environment. The course material to be learned is only one of the four components of learning. This is essentially what podcasting is about. Podcasting describes the manner in which the course content is packaged and delivered to students. The other three components are also related to podcasting as teaching and learning tools. The environment component is related to the flexible learning environment. Boettcher suggested three ways to design course content: (a) Content must be semantically well structured for instruction, (b) content must be a good fit or well structured for a particular student, and (c) content must be technologically well structured. The first way corresponds to the teaching component of the learning experience, the second way corresponds to the learner component of the learning experience, and the third way corresponds to the environmental component of the learning experience.

Podcasting is technology-based learning. It is a structured learning process in which course content is tailored using all of Boettcher's three ways. These findings support the use of podcasting as teaching and learning tools and illuminates why podcasting is a "good fit" for today's technological environment (Lee et al., 2008,

pp. 532-533). Boettcher's study was also supported by American psychologist Jerome Bruner's research on how students understood "the conceptual underpinnings of knowledge." Bruner argued that learning depends on environmental and experiential factors. "The way in which instruction was organized and what was emphasized during instruction were keys to improving learning" (as cited in Salkind & Rasmussen, 2008, p. 964).

The Impact of Technology on Teaching and Learning

According to Gumport and Chun (2005), there are three potential influences of technology on higher education: (a) the nature of knowledge, (b) the process of teaching and learning, and (c) the social organization of teaching and learning in higher education. The nature of knowledge is determined by how people are involved in the production of knowledge and how academic knowledge is valued and disseminated. The process of teaching is the process in which teachers and students come together in the same place at the same time for the dissemination of knowledge. The third potential influence is the social organization of teaching and learning in higher education that is determined by expanding the delivery of higher education (pp. 402-412). This idea suggests that technology can be used to extend the traditional teaching and learning process without altering the underlying the traditional forms of education.

Traditionally, teachers and students come together in a classroom learning environment for the dissemination of knowledge. Based on the work of Cornford and Pollock, "Working through the Work of Making Work Mobile," Robins and Webster (2002) stated that "the emergence of commercially-oriented, digital, online or virtual university" allowed higher educational institutions to create new and different ways of producing, distributing, and consuming knowledge that learning does not have to be

“campus-based” (p. 87). With educational podcasting, knowledge is packaged and information is exchanged between teachers and students; teachers are the producers and students are the consumers of knowledge. Teachers and students do not have to be at the same place at the same time, where a physical classroom is a virtual place for teaching and learning (Zajda & Gibbs, 2009, p. 21).

Well-known technologist Bill Gates spoke about “the Web lifestyle” as a catalyst of change in collaborative learning communities in which the Internet allows teacher and students to explore a subject matter in a new ways and learning does not have to be in a classroom learning environment (Gates, 1999, p. 387). Today’s teaching and learning are no longer restricted to teachers and students coming together in the same place at the same time, and a physical classroom is not the only place for teaching and learning; they can take place in both traditional classroom education and distance education (Manca, 2008, pp. 121-122). Like the Internet and Web technologies creating opportunities for distance education, e-learning and online education provides anywhere and anytime learning for students without constraint of a physical classroom.

Many American universities and academic institutions around the world have utilized technology and Web-mediated learning for academic training. Classroom education has blended the use of technology to facilitate teaching and learning through use of computer, projector, presentation software applications, the Internet, multi-media, and Web-based course tools in classroom learning environments (Ng, 2008, pp. 21-29). Current trends in mobile technology have expanded teaching and learning from current traditional classroom education and distance education to flexible teaching and learning practices.

Distance education such as online and e-learning use the Internet and Web-mediated technologies to facilitate traditional classroom learning. Academic institutions around the world have adopted these technologies to create distance learning opportunities for their students and to promote interaction among students and between teachers and students (Ng, 2008, p. 21). Several popular Web-based course management systems for students include Blackboard, TopClass, and government-sponsored Moodle. These learning systems used Web-mediated technologies to provide a learning environment that is alternative to traditional classroom settings. In addition, these learning systems allow flexible learning environments for students, such as study from home, at their own pace, and while they maintain a personal life or job. The International Technology Education Association (ISTE) defined educational technology as “using multimedia technologies or audiovisual aids as a tool to enhance the teaching and learning process” (Salkind & Rasmussen, 2008, p. 314). These learning systems are closely aligned with ISTE. According to Driscoll (1998, as cited in Donnelly & McSweeney, 2008, pp. 51-52), a Web-based learning platform provides a complete analysis of all types of asynchronous interactions between teachers and students, students and students, and students and course material. Current technological trends in mobile learning have created flexible learning methods for teachers and students to effectively communicate, interact, and engage in classroom learning activities that complement traditional classroom and distance education.

Traditional classroom and distance education, and now mobile teaching and learning have created new opportunities for mobile learning or m-learning (Kim et al., 2006). Business, education, and people worldwide have increasingly relied on mobile technology for everyday purposes. As the Internet has dominated the world of information, mobile technology is a new way to access the Internet. Mobile devices

such as cell phones, PDAs, and related mobile devices are much more than just handheld computer systems; they are multi-purpose devices for communication, accessing the Internet, PDA, and educational purposes. In recent years, innovation in mobile technologies has allowed mobile devices to access distance learning, distributed learning, e-learning, and now m-learning (Frignani, 2008, p. 171). These devices create new opportunities for flexible learning in addition to communication, collaboration, and exchange information (Kidd, 2008, pp. 847-851). In this context, mobile technology has reshaped classroom learning and distance learning. Mobile learning has given a new meaning to current distance education.

The pedagogical approaches to podcast-mobile-assisted learning combine both information technology and traditional classroom teaching and learning to provide the same learning experience as traditional classroom education. Commonly used online discussion groups, videoconferencing, and online resources in distance education are now incorporated in m-learning (Hussain, 2008, p. 69; Reid-Martinez et al., 2008, p. 703; Vanin et al., 2008, p. 1). Several studies have shown that teachers have more positive attitudes toward the potential of technology-based education and teachers who lacked technology experience had negative attitudes toward computers and did not use technology in their teaching (Jimoyiannis, 2008b, p. 325). For this reason, the future of higher education must be directed to the training of teachers in adapting to the new learning environment and becoming more integrated with technology.

The Use of Podcasting for Teaching and Learning

As Internet and Web technologies have created distance education and online learning environments, innovations in podcasting have added a new dimension of mobile learning to the traditional forms of classroom and distance education (Song &

Kidd, 2008, p. 569). Mobile devices such as cell phones, PDAs, and related media devices have enabled delivery of classroom education through the process of podcasting. Podcasting is simply the process of packaging classroom learning activities into audio/video podcast files and distributing them over the Internet to students' mobile devices for playback. The packaging process is the recording of classroom learning activities into audio/video podcast (or MP3/MP4) files and posting them on faculty Web sites or Apple's iTunes software application. Learners access, download, and playback audio/video podcasts of the learning activities that take place in the classroom. Learners access and review lessons covered in the classroom, listening to them once or multiple times. With audio/video podcasts, learners are in total control of their own learning, without time or geographic constraints. Unlike distance learning, podcasting is a prerecording of audio/video podcast of classroom learning activities. Like distance learning, podcasting learning is a flexible learning process beyond a physical classroom and geographic location, and the learners still have the same classroom learning experience. Jonassen (2006) described that the role of technology is to "communicate ideas to learners" (p. xiii). Podcasting is essentially a teaching/learning tool that has the capacity to deliver classroom learning activities via mobile devices.

Podcasting is a supplement to classroom instruction, essentially a recording of class lectures and reviewing them outside the classroom. This approach is not a new concept, but podcasting of class lectures via mobile devices for students at the community college is a new concept. According to Salkind and Rasmussen (2008), educational technology started around the 1950s and focused primarily on the use of instructional media. However, the use of audio recordings for listening started as early as the 1920s (p. 314). An exploratory study using tape-recorded journals by Aitken and Deaker (2007) concluded that the use of tape recording was an effective means of

expression for some students and an important aid to critical thinking. Today's podcasting has now emerged as a new tape recording and is evolving as the new evolution of educational technology.

Before understanding the benefits of podcasting in educational settings or being accepted as teaching/learning tools, it is prudent to understand what it is. In this way, actions can be taken to determine whether podcasting is an effective teaching/learning tool for today's technological society. The term *podcasting* is derived from Apple's portable media players, iPods. Podcasting technologies have the capability to package and deliver multimedia content in podcast file format (or MP3/MP4). Learners can subscribe, access, download, and play back audio/video podcasts on iPods, MP3/MP4 players, iTunes (Apple's media software player), Windows Media Player, or related mobile devices.

In technical terms, podcasting is an audio/video files feed, in Really Simple Syndication (RSS) feed format, which is a family of Web feed formats used to publish frequently updated digital content. The term *podcasting* is also used to describe a collection of technologies for automatically distributing audio/video content over the Internet. Educational podcasting can be described as the recording and distribution of the instructional materials such as lectures, class activities, and assignments in digital format or MP3/MP4 format. Windows Media Player, iTunes software application, or related media players can play back audio/video podcasts. However, the term *podcast* still refers largely to audio/video content distribution. Audio/video podcast is either in "push and pull" or audio/video files feed through RSS. A "push and pull" is essentially an audio/video podcast in MP3/MP4 that users can download. An audio/video files feed is automatically synchronized with the users' mobile devices.

In educational terms, podcasting is simply a process for teachers to create self-published course content and distribute it as podcast files. Educational strategies have two aims: (a) enhancing the method of teaching, and (b) helping students to learn more effectively. First, the teacher can listen and improve teaching and presentation skills by listening to his/her own lectures and classroom learning activities. Second, students can download and play back any of the published classroom podcasts of the learning lessons that are covered in the classroom. Students can listen to or review any of the lectures or related learning lessons outside the classroom. The pedagogical model is a “listen and learn” learning concept that students can play back audio/video podcasts of the learning lessons on their mobile devices. With podcasting, students may choose to listen once or multiple times to the same lessons. When students actively engage in the learning lesson by listening to the same podcast multiple times, they have constructed knowledge and the information that they have collectively accumulated by listening and beginning to crystallize the information. This idea is similar to the work of Herasymowych on increasing intelligence through actions that reflect on the learning cycle, action learning, and learnable intelligence (Merriam & Cafarella, 1999, pp. 188-189).

Mobile technologies and podcasting are tools to help students to think more effectively. As a teaching tool, podcasting works like learning lessons as tutorials on mobile devices that provide step-by-step instructions to help students to construct the meaning of the learning lessons. Teachers provide the learning lessons through audio/video podcasts that playback on students’ iPods, MP3/MP4, cell phones, or related media players. Traditionally, learners attend lectures to get clarification on learning topics. This teaching method is simply building on the success of the Internet and Web technologies that aim to create new opportunities for faculty and students to

collaborate and engage in classroom teaching in a new way that provides the ability for faculty to create and relay course materials as audio/video podcasts. By combining traditional classroom method, the Internet and mobile devices allow teachers to construct effective learning lessons in audio/video podcasts. According to Kim et al. (2006), learning through mobile devices is a new form of m-learning with no limitation on the place or the time.

The learning process of podcasting utilizes the Internet to download either through “push and pull” or RSS of audio/video podcasts and playback on students’ mobile devices or related media players. This approach combines both classroom and distance learning (i.e., online and e-learning) to create m-learning. As Pieri and Diamantini (2008) have stated, e-learning is “the natural evolution” (p. 548). This pedagogical approach of podcasting will not replace traditional classroom experience; it simply reinforces classroom learning and offers students flexibility to learn in a technological culture with which they are comfortable. This approach utilizes mobile technology and the Internet to distribute classroom education to mobile devices. Podcasting learning offers a flexible way of learning other than traditional classroom education. This form of education is currently very appealing to working students, because podcasting provides flexible learning that fits the learning preferences and culture of students (Inoue, 2008, p. 19). Podcasting aims to provide students additional time to review classroom learning activities that are previously covered in classroom and still have the same learning experience as in the classroom.

Learning Theories and Practices

The art of teaching is the shaping of individual lives to influence their integration into productive societies; it requires blending theory and practice. Practice is

guided by proven theories (Gutek, 1997, p. 1). A philosophy of education is simple: to help a person to understand reliably the social tenets of reality, the essence of truth, and values.

There are essentially three branches of philosophy: metaphysics, epistemology, and axiology. Metaphysics deals with social realities. Epistemology is the essence of truth and the theory of knowing and knowledge. Axiology is concerned with value-based theories (Gutek, 1997, pp. 2-3). These fundamental branches of philosophy help to determine whether the use of mobile centric and podcasting devices advances the field of education significantly or materially. Three basic concerns are addressed here: (a) theorizing the practice of mobile and podcasting technologies, (b) the theory of knowing and knowledge society, and (c) the societal value of mobile and podcasting technologies.

Based on educational learning theories and the nature of reality, the pedagogical practice of podcasting is a combination of constructivism, behaviorism, social learning theory, and cognitive theory. Behaviorism presupposes that students actively engage in self-directed learning, social learning theory presupposes that students learn through the help of audio/video podcasts, and cognitive theory presupposes that students learn through building on prior experiences and through meaningful and relevant experiences (Merriam & Cafarella, 1999, pp. 263-266; Wong & Kerr, 2008, p. 40). Constructivism learning theory suggests that students learn when they actively construct knowledge from listening to classroom podcasts (p. 182). Social learning theory is considered to be one of the most influential theories in the fields of education and psychology. Social learning theory in terms of human functioning and personality development is linked to culture, society, and community. The theory is based on cognitive, social-interactive, self-regulatory, and self-reflective capabilities and

processes and concludes that social network communities have a great influence on human learning and development (p. 919). In addition to the impact of social interactions, social learning theorists believe that media have a great influence on human learning and development. These theories suggest that teachers and students are closely related to cultural values and learning styles (Salkind & Rasmussen, 2008, p. 152). Based on these theories, mobile and podcasting technologies can be used to create the sharing of culture, embrace the technological society, and enable teachers and students to collaborate. The relationship between theory and practice is to innovate ways to adjust to specific learning situations. The nature of reality is that society embraces the evolution of technological changes that drive innovation and new ideas. According to Smith, the nature of reality and the essence of truth are “true to experience” and reflect the “real world” (2003, as cited in Curren, 2003, p. 384).

Podcasting learning can be based on perceptual development. It is consistent with constructivism, behaviorism, social learning theory, and cognitive theory. Salkind and Rasmussen (2008) mentioned two perceptual developments of senses: sensations and perceptions. Sensations are about information received from the environment, such as a visual feature or a sound. Perceptions are organized and interpreted sensations, such as recognizing a face or interpreting a sequence of sounds as a familiar song (p. 774). Podcasting is related to hearing and touching of the four senses (hearing, smelling, tasting, and touching). Listening is the hearing of audio/video podcasts, and controlling the speed of “listen and learn” is touching. Cognitive developmental theorist Jean Piaget (1896-1980) argued that “the knowledge and skill underlying intermodal perception can only be gained through experiences that involve looking at, hearing, smelling, tasting, and touching the external world” (p. 778).

Adapting to these theories, environmental and technological changes are commonly adapted by society. The Internet and innovations in Web technologies have breathed life into the knowledge society. Innovation in mobile learning and podcasting allow people to communicate and exchange information in today's social structure. The question remains, "What is the best innovative way to learn"? One answer is to find a suitable way to adapt to today's realities of students and educators. In this context, education through podcasting is new teaching/learning methodology for new realities of the learning styles of today's academia. The new generation of college students already has extensive experience with the Internet, Web-enabled devices, and mobile technologies; and these technologies are fast becoming a significant part of their daily lives (Gallerani, 2008, p. 19). Due to social change, people are more readily accepting move toward the Internet, the Web, and mobile technology-based teaching/learning. The Internet and Web technologies are indispensable elements in today's society. *San Jose Mercury News* columnist Mike Cassidy (2009) wrote a column called "Lessons Learned from a Tech-Free Day." He asked teenagers from the local high school to refrain from using their mobile devices for 12 hours: no iPods, no cell phones, no PDAs, no video games, no DVDs, no TVs, no PCs. Even though the "tech-free day" was merely a gesture, it showed that new generations of students are enthusiastic about their mobile devices.

Building on the successes of the Internet, Web-enabled technologies and mobile technologies offer a wide range of disciplines including business, education, and entertainment. This phenomenon opens unlimited possibilities for the way people live, learn, and work (Sorrentino, 2008b, pp. 215-216). It is a common behavior to learn and adapt to the information society, and the mobile society is a natural transition (Merriam & Cafarella, 1999, pp. 73-77). Consequently, higher educational

institutions and global industries need to form partnerships to enhance the learning styles of students. Bill Gates (1999), founder Microsoft, the world's most successful computer company, stated, "The Web lifestyle isn't about changing human nature of the fundamentals of how people live. Instead the Web lifestyle gives people a chance to pursue their interests in a better way" (p. 131). Emerging mobile and podcasting technologies are just a natural evolution of "Web lifestyle" and beyond. The theory of knowing and knowledge society is to find the most appropriate way to learn and to be successful. This essentially means that, if the use of podcasting helps students in cultivating reason in the "real world," then podcasting has achieved its purpose.

Mobile technologies have changed greatly since their inception. Today's mobile computing can be in the form of a personal computer, PDA, wireless networking, cell phone, iPod, any of the home network appliances, or any other digital electronic device. These digital devices are becoming an indispensable part of people's lives, and they are responsible for connecting schools, businesses, and people from virtually anywhere in the world. Society changes force people to apply learning in a more pragmatic approach to education, and this essentially is what podcasting entails.

In terms of the value of mobile and podcasting technology to society, podcasting provides numerous ways for students to learn, and in effect, transforms classroom learning as we know it today. According to John Dewey (1859-1952), American pragmatic philosopher and proponent of experimentalism in education, students learn by getting involved in play, construction, nature study, and self-expression (Gutek, 1997, p. 82). Dewey's school and classroom approaches were conceived as an embryonic community in which students work together, in common, to solve mutually shared problems (p. 96).

Podcasting is also consistent with cooperative learning and Deutsch's work on social psychological theory, which suggests that the social and motivational environment promotes positive and supportive peer interactions and a positive orientation toward achievement and learning (Salkind & Rasmussen, 2008, p. 188). If the effective use of podcasting improves students' classroom experiences, then there are values added to the field of education, especially the teaching and learning environments. Based on what is known about podcasting, podcasting can be an effective tool to facilitate collaboration and information exchange. Podcasting learning concept is very similar to cooperative learning, which suggests that the social and motivational environment in the classroom promotes positive and supportive peer interactions and a positive orientation toward achievement and learning (Salkind & Rasmussen, pp. 187-188).

Podcasting learning theory and practices are simply a concept of "listen and learn." These educational practices provide opportunities for today's lifestyles and learning styles of students. This pedagogical approach is consistent with the constructivism approach to self-directed learning that suggests that people construct new knowledge from their past experiences and that the curriculum and instruction should be tailored to facilitate development of individual perfections (Ferri, 2008, p. 82; Merriam & Caffarella, 1999, p. 69; Wong & Kerr, 2008, p. 42). Emerging podcasting methods for teaching and learning are based on the idea of anytime and anyplace learning for today's lifestyles.

In terms of learning styles, learners learn best in their environment, namely learning styles and learning settings. According to Salkind and Rasmussen (2008), teachers should use learning style information to plan teaching strategies, learning activities, evaluation of learning, and development of programs and curricula (p. 598).

According to Ned Herrmann (1922, as cited in Salkind & Rasmussen, pp. 600-601), there are four learning styles: (a) theorists (rational), (b) organizers (safe keeping), (c) innovators (experimental), and (d) humanitarians (feelings). Podcasting learning theory and practices are based on the idea of flexible learning styles of today's life-styles of students.

Podcasting learning works like the traditional form of taking notes and reviewing classroom lecture notes. Traditionally, taking notes is a common activity in a classroom setting, and a large percentage of students are regular note takers (Hussain, 2008, p. 69; Palmatier & Bennett, 1974). Several studies have investigated the art of effective note taking. According to Cohn, Cohn, and Bradley (1995), students must actively pay attention to the lecturer while listening and taking notes, and then hold and manipulate a representation of that information in working memory while recording notes. As a consequence, students are less likely to pay attention to the lecture while taking notes. In another study, Kreiner (1997) found that interactive videotape was more effective than taking notes while watching videotape, because interactive videotape allowed students to focus on relevant concepts and encouraged classroom participation. Similarly, podcasting offers flexible interactive learning outside the classroom and allows students to pay attention to the lecture.

Kiewra et al. (1991) delineated suggestions to facilitate the selection of information to be recorded during a lecture. Cueing strategies, such as recording information on the blackboard, repeating the lecture, and use of a tape or video recording, can be used repeatedly to enhance student learning. Classroom podcasts are essentially audio/video recordings of class notes, lecture notes, and cue cards that entail the detailed summary, outline, or tutorial of the learning lessons. Fryer (2008) found that technology-assisted learning had positive effect on lectures, especially in

taking notes and study for examinations. Smith and Woody (2000) found that technology-facilitated instructions were perceived more favorable by students than were traditional classroom instructions. Using podcasting technology to create highly participative, rich learning experiences, review classroom learning materials by reinforcing students' rapt attention, and respond more quickly to changes in environments enabled learning, fostering retention and learning more effectively. Based on these findings, a technology-mediated approach significantly improved the student learning process and outcomes as measured by student self-report and objective outcome testing. A survey conducted by the Berenson Strategy Group for Common Sense Media (2009) found that 41% of the students (7th to 12th graders) reported that they stored class notes on their cell phones.

With educational podcasting, the Internet, and Web technologies have created information-rich teaching and learning environments that enable social interaction and exchange of information between teachers and students. The Internet and Web technologies have allowed teachers and students to gain access to a wealth of knowledge-based information. Podcasting-supplemented learning tools can be used to replicate and expand the traditional classroom experience. Similarly, podcasting can be used to widen access and retrieve and disseminate knowledge by providing anytime and any-place learning (Aggarwal & Bento, 2000).

Today, technology is used in virtually all American schools (K-12, colleges, and universities). Technology is integrated into the classroom core curriculum, making classroom learning more accessible. By reaching out to the world through the Internet and now mobile (or m-learning), the experience can be extended beyond the traditional forms of instruction (Lastrucci, Infante, & Pascale, 2008, p. 160). Weaving

the popular Internet communication service and technological culture of students, academic institutions may be able to engage and motivate students to learn.

The emerging mobile and portable technologies in cell phones, PDAs, and MP3 devices make podcasting possible. With podcasting, classroom instruction materials can be broadcast into digital audio/video learning modules that students can download in order to learn the presented materials. Today's podcasting is become ubiquitous to describe the packaging and distributing of audio/video contents over the Internet. A definition of podcasting offered by Merriam-Webster is "a program (as of music or talk) made available in digital format for automatic download over the Internet" (Merriam-Webster, 2001, p. 1). Innovations in podcasting have significantly changed teaching and learning in college education. With podcasting, the Internet, mobile and portable media devices, classroom instruction can be broadcast in RSS feed format and downloaded on demand. Like the Internet, mobile and portable media technologies provide greater flexibility in the learning process and are revitalizing today's college education (Reid-Martinez et al., 2008, p. 703).

Emerging podcasting technology makes it possible for community college students to access classroom learning materials on demand. This learning technique simply incorporates classroom experiences into students' own culture of technological identity to motivate them to learn the subject matter. Podcasting will not replace traditional classroom learning experiences but rather will reinforce students' learning appetite. Several researchers have suggested that when students are motivated to learn in their technological identity environments, they are more eager to learn the subject matter, perform better on tests, and ultimately achieve better grades (Ranieri, 2008, p. 318). A study at the Catholic University Faculty of Economics, technology-mediated learning showed significantly higher mathematics scores in students'

achievement tests (Torriero & Scancarello, 2008, p. 263). Dettori and Persico (2008) found that motivation was a key element in students' academic performance (p. 741). As O'Banion (1997) stated, "One can no longer afford to ignore technology and still maintain institutional health" (p. 64). Podcasting has become an indispensable part of the students' technological identity.

These technology trends will continue to provide new methods to help people to interact, businesses to extend commerce, and governments to empower citizens and advance societal virtues. Based on trends in technology, there is potential for a convergence of business, education, and technology in knowledge sharing. The vision of the future in technologies will likely be based on centralized knowledge-based systems that link knowledge databases. According to Motwani and Akbulut (2008), there will be more virtual classrooms in the future of educational training (p. 305). Given these results, growing numbers of teachers have already started using technology media outlets and related technology-enabled instruction in their classroom learning environment.

Given the findings from these studies, educators will have a better understanding of students' use of technologies relative to traditional classroom methods of teaching. Educational podcasts are a relatively new technology that allows online audio/video between students and teachers, delivered via RSS podcasting that can be listened to on students' PDAs or personal computers. In lieu of reviewing cryptic class notes, students have the opportunity to replay class lectures via the podcasting process. This also comes with a cost for both students and teachers, in that they must have the iPods, the iPhones, or related MP3/MP4 players. In addition, teachers must regularly publish podcasts of the class materials.

In conclusion, there are many reasons for adapting to the real world of technological and societal changes. The general idea is to adapt, create, educate, empower, and innovate ways to enhance students' learning outcomes. By adapting to mainstream culture, technological changes, lifestyles, and learning styles of students, higher educational institutions can confront these changes by revitalizing "the old schools" thinking and create new ones that are in line with the real world of technological changes and societal changes (Goethals, Sorenson, & Burns, 2004, pp. 1134-1144).

In order to integrate podcasting effectively in the classroom teaching/learning environment, support effective training courses, and make the learning more effective, future leaders of higher education must have a plan for societal changes and be ready to respond to technological challenges, especially with the younger generation of students who have grown up in the digital world. They must identify, recruit, and train staff members for leadership positions that help them to build more integrated ways to teaching and learning that will empower students to take control of their own learning and use the learning tools available to achieve desired learning outcomes. That includes training teachers to use technology to enhance their teaching methodology and adapt to their personal development and enable them to become more efficient and effective teachers. The technology trends discussed have shown that not only business enterprises but also higher educational institutions have been greatly affected by the advent of the Internet and now mobile technologies in cell phones, PDAs, and related media devices. These innovations have virtually transformed every aspect of society, including higher education, business, government, and global industries. Many of these institutions have already undergone paradigm shifts and major repercussions with respect to restructuring to meet the needs of all of their stakeholders.

For these reasons, future leaders of higher education must find new ways to overcome the digital divide between teachers and students, specifically the students who are academically underprepared and economically disadvantaged. In addition, they must put the needs of their stakeholders in the forefront and adapt to societal and technological changes to reach out to future students. After all, the real purpose of education is to educate students. The challenge is not so much about podcasting and its ability to package/distribute course content as digital content; rather, it is to ensure that quality teaching meets the needs of the students and satisfies the learning requirements by embracing societal and technological changes without derailing the purpose of the institution.

Podcasting is a new way of teaching and interaction between learners and knowledge. As Bill Gates (1999) pointed out, “We always overestimate the change that will occur in two years and underestimate the change that will occur in the next ten” (p. 69). For these reasons, understanding the role of the technological culture of students and the technological society is a must for future leaders of higher education.

WSIS acknowledged that it is a global responsibility, including higher education, to “create access, utilize and share information and knowledge, enabling individuals, communities and peoples to achieve their full potential” (WSIS, 2003, para. 1). For this reason, higher institutions of learning must reconcile the importance of technology-based teaching/learning to the overall success of the institutions’ practices.

Chapter 3

METHODOLOGY

A quasi-experimental design was used to study the effect of classroom podcasts in regard to the average midterm and final examination scores of students. Chapter 3 describes the methods and procedures involved in this study. The chapter is organized in the following sections: (a) Restatement of the Problem, (b) Hypotheses, (c) Research Design, (d) Population, (e) Data Sources, (f) Validity and Reliability, (g) Limitations of the Study, (h) Statistical Analysis, and (i) Summary.

Restatement of the Problem

Podcasting is a dynamically evolving technology that outlines the manner in which audio and/or video content may be accessed from cell phones, portable media players, and personal computers. According to Brown and Green (2007-2008) and Podcasting News (2008), podcasting is considered to be the fastest-growing technology and it appears to be growing exponentially. Recent advances in podcasting presentations and digital media capabilities have expanded from personal entertainment usage to educational purposes.

With rapid growth and the adaptation of mobile and podcasting technologies in the general population and higher education, podcasting has fundamentally changed the classroom teaching and learning environment. Many academic institutions, including Duke University, Harvard University, MIT, and Stanford University offer course

materials via audio/video podcasts (Chan et al., 2006; Fryer, 2008). *The Chronicle of Higher Education* (Read, 2005) reported that in 2004 Duke University distributed iPods to all first-year students as part of an initiative to encourage the use of podcasting in education. After evaluating the program, Duke University revolutionized the adoption of the Pods or simply MP3 media players as a course content dissemination tool. Not only is podcasting being offered nationally at leading universities; Ohlone College, the site for this study, launched Ohlone's iTunesU, aimed to provide capability for students to access and download audio and video podcasts to their computers or iPods directly from iTunes (Ohlone College, 2006).

Ohlone College, like most of American community colleges, is geared toward nontraditional students (adult learners and lifelong learners) rather than traditional college students in 4-year colleges and universities. According to a report on student demographics by the National Commission on Community Colleges (2008), the average age of students at Ohlone was 29; 43% of the students were in the 21-or-younger age group and 42% were in the 22-39 age group; the 40-or-older age group was 16%; 59% of the students were women, 34% were minorities, and 8% were non-U.S. citizens. Traditional college students are typically in the 18-24 age group and nontraditional students are in the 25-and-over age group. Furthermore, community colleges serve students who are academically underprepared, economically disadvantaged, the first in their family to attend postsecondary education, or members of traditionally low-performing subgroups. In addition, there is a digital gap between traditional and nontraditional students.

As reported by the National Commission on Community Colleges (2008), The Education Commission of the States (2008), and Ohlone College (2009), the data suggest growing numbers of nontraditional students enrolled in American community

colleges. The data showed that traditional college students also enrolled in community colleges in large numbers. The National Center for Education Statistics (2007) projected a rise of 15% in enrollments of people under 25 and a rise of 21% in enrollments of people 25 and over by 2016. These data suggest that nontraditional students are returning to school at a rapid rate. Ohlone College, like most of the California community colleges, serves students affected by the digital divide. Podcasting is just one of the ways to overcome the digital divide between traditional and nontraditional students, especially economically disadvantaged students.

Podcasting is simply the process of packaging/distributing training course contents into audio/video podcasts. Podcasting allows faculty to create, publish, and post course content on Web sites (Barlow, 2008; Mikat et al., 2007). Students can access and download classroom podcasts either from faculty Web sites or from Ohlone's iTunesU, to play back on their cell phones, iPods, iPhones, MP3/MP4 players, or personal computers. The additional time to review classroom lectures, lessons, and assignments afforded by classroom podcasts offers students an alternative reinforcement for learning key concepts for the course. This method is an effective way for educators to communicate curriculum and other vital information to students without the constraint of the classroom (Kui & Mengmeng, 2007). According to Barlow (2008) and Fryer (2008), students were likely to be more engaged in subject matter when they used podcasts because they found that podcasts made learning fun and exciting.

Not everyone agrees that podcasting is an effective tool for the classroom. Some educators maintain that podcast recording of lecture materials may not be effective in facilitating classroom learning (Copley, 2007; Farivar, 2007). Copley conducted a pilot study on the production of audio and video podcasts of lectures for

campus-based students. This pilot study indicated that podcast recordings of lectures were not effective in facilitating mobile learning. As Frydenberg (2006) pointed out, teacher/student relationships and classroom attendance may have been impacted by the broad availability of podcasts because students were able to download classroom materials without having to attend classes.

Although learning through podcasting methods offers new flexible learning outside the classroom, it also presents challenges to traditional on-ground classroom learning. One of the major challenges facing classroom learning is how to engage the diverse learning styles of student, especially community college students (Lyles et al., 2007). With the flexibility of classroom podcast learning, students choose their preferred learning styles to succeed in their overall course work. However, the benefits of incorporating classroom podcasts into traditional classroom settings remain unclear.

In an effort to address these concerns and the effectiveness of podcasting in the classroom teaching and learning settings in Ohlone College, this study aimed at investigating classroom podcast usage by community college students and their test scores. The supposition was that students who listened to classroom podcasts would be more likely to achieve higher test scores than students who did not listen to classroom podcasts. In addition, this study addressed the use of classroom podcasts to reinforce student learning and strengthen classroom instruction. This research study focused on the following questions:

1. Are there significant differences between two types of educational instruction (traditional classroom instruction and classroom instruction supplemented with classroom podcasts) in regard to the average midterm and final examination scores of students?

2. Is there a significant difference between the numbers of times students listen to classroom podcasts before and after the midterm?

3. Is there a significant difference between class attendance (less than other classes, same as other classes, and more than other classes) with regard to the number of hours students listen to classroom podcasts per week?

4. Is there a significant difference between age groups with regard to the number of hours students listen to classroom podcasts per week?

5. Is there a significant difference between genders with regard to the number of hours students listen to classroom podcasts per week?

Research Hypotheses

To address the research questions, hypotheses were constructed for each of the research questions. The hypotheses were tested on the overall classroom podcast usage by community college students and their test scores. For each of the questions there were two hypotheses: the null hypothesis (H_0), and the alternative hypothesis (H_a). If the null hypothesis yields no evidence to suggest a difference, then no further test is needed. When the null hypothesis shows a statistically significant difference between the frequency of the overall usage in relation to the students' test scores, then the null hypothesis is rejected that the alternative hypothesis is accepted (Freedman, Pisani, Purves, & Adhikari, 1991, p. 432; Levin & Stephan, 2005, pp. 125-132). That is, if results from the statistical tests show no evidence to suggest a difference between the two types of classroom instruction (traditional classroom instruction and classroom instruction supplemented with podcasts) and the students' final test scores, then no further test is needed.

Research question 1: Are there significant differences between two types of educational instructions (traditional classroom instruction and classroom instruction supplemented with classroom podcasts) in regard to the average midterm and final examination test scores of the students?

H₀1: There are no differences between two types of educational instructions (traditional classroom instruction and classroom instruction supplemented with classroom podcasts) and the average midterm and final examination test scores of the students.

H_a1: There are differences between two types of educational instructions (traditional classroom instruction and classroom instruction supplemented with classroom podcasts) and the average midterm and final examination test scores of the students.

Data: The midterm and final examination test scores from two types of educational instruction (traditional classroom instruction and classroom instruction supplemented with classroom podcasts) were obtained from students who experienced traditional classroom instruction and classroom instruction supplemented with classroom podcasts in their classroom instructions over the course of an academic semester.

Research question 2: Is there a significant difference between the numbers of times students listen to classroom podcasts before and after the midterm?

H₀2: There is no significant difference between the number of times students listen to classroom podcasts before the midterm and after the midterm.

H_a2: There is a significant difference between the number of times students listen to classroom podcasts before the midterm and after the midterm.

Data: The numbers of times that students listen to classroom podcasts were obtained from students before and after the midterm.

Research question 3: Is there a significant difference between the class attendance (less than other classes, same as other classes, and more than other classes) on the number of hours students listen to classroom podcasts?

H₀₃: There is no significant difference between class attendance (less than other classes, same as other classes, and more than other classes) on the number of hours students listen to classroom podcasts.

H_{a3}: There is a significant difference between class attendance (less than other classes, same as other classes, and more than other classes) on the number of hours students listen to classroom podcasts.

Data: The number of hours that students listen to classroom podcasts (1 hour or less, 2 hours, 3 hours or more, and not at all) and the average class attendance were obtained from students who experienced traditional classroom instruction and students who experienced classroom instruction supplemented with classroom podcasts over the course of an academic semester.

Research question 4: Is there a significant difference between age groups with regard to the number of hours students listen to classroom podcasts?

H₀₄: There is no significant difference between age groups with regard to the number of hours students listens to classroom podcasts.

H_{a4}: There is a significant difference between age groups with regard to the number of hours students listen to classroom podcasts (1 hour or less, 2 hours, 3 hours, 4 hours or more, and not at all).

Data: The number of hours that students listened to classroom podcasts (1 hour or less, 2 hours, 3 hours, 4 hours or more, and not at all) and age groups (19 years or younger, 20 to 24 years, 25 to 29 years, 30 to 34 years, 35 to 39 years, 40 to 49 years, and 50 years or older) were obtained from students who experienced traditional

classroom instruction and students who experienced classroom instruction supplemented with classroom podcasts over the course of an academic semester.

Research question 5: Is there a significant difference between genders on the number of hours students listen to classroom podcasts?

H₀₅: There is no significant difference between genders on the number of hours students listen to classroom podcasts.

H_{a5}: There is a significant difference between genders on the number of hours students listen to classroom podcasts.

Data: The number of hours students listened to classroom podcasts (1 hour or less, 2 hours, 3 hours, 4 hours or more, and not at all) and gender status were obtained from students who experienced traditional classroom instruction and students who experienced classroom instruction supplemented with classroom podcasts over the course of an academic semester.

Research Design

A quasi-experimental design was used to determine the effect of classroom podcasts in regard to the average midterm and final examination test scores of the students. The design involved students who experienced two types of educational instructions (traditional educational instruction and classroom instruction supplemented with classroom podcasts) in their classroom instruction over the course of an academic semester. Traditional classroom instruction (attending lectures, reading the textbook, and taking notes) was used as a control. The treatment (experimental) group consisted of students who used the supplemental classroom instructions with podcasts. The control group consisted of students who used only traditional classroom instruction (Freedman et al., 1991, pp. 3-10). Midterm and final examination test scores were

used to determine the significant differences between the control group and the treatment group. Scores from the two groups were compared to determine the effect of classroom podcasts in regard to their midterm and final examination test scores. Age groups, classroom attendance, ethnicity, and gender of students were also examined to determine within-group (classroom instruction supplemented with classroom podcasts) differences. According to Salkind and Rasmussen (2008), this design is an appropriate procedure when studying two intact groups.

A two-group nonequivalent control group design was used to study the differences between two types of educational instruction (traditional classroom instruction and classroom instruction supplemented with classroom podcasts). The posttests between two types of educational instruction were used to determine the effectiveness of classroom instruction supplemented with classroom podcasts (treatment group; Gay, Mills, & Airasian, 2006, pp. 255-258). First, two types of educational instruction were administered with the midterm test. Second, the same treatment applied to midterm was posttested for the final examination to strengthen the generalizability of the findings (Figure 1).

The procedures entail the production, publication, and delivery of audio podcast. Production is the recording of the course contents as audio podcasts by the course instructor. Publication is the posting of audio/video podcasts on the instructor's Web site or Ohlone iTunesU. To ensure compatibility between iPods and MP3 players, students may choose to use Apple's iTunes application to synch the MP3 files or download the files directly from the instructor's Web site or related Web-based course tools. Delivery is the distribution of audio/video podcasts from the instructor's Web site or Ohlone iTunesU to the students' cell phones, PDAs, MP3 players, or personal computers (Figure 2).

Nonequivalent Control Group Design	X1 (midterm)	O	X1 (final exam)	O
	X2 (midterm)	O	X2 (final exam)	O

Figure 1. Grouping of participants. X1 = classroom podcasting (experimental treatment), X2 = no classroom podcasting (control treatment), O = posttest.

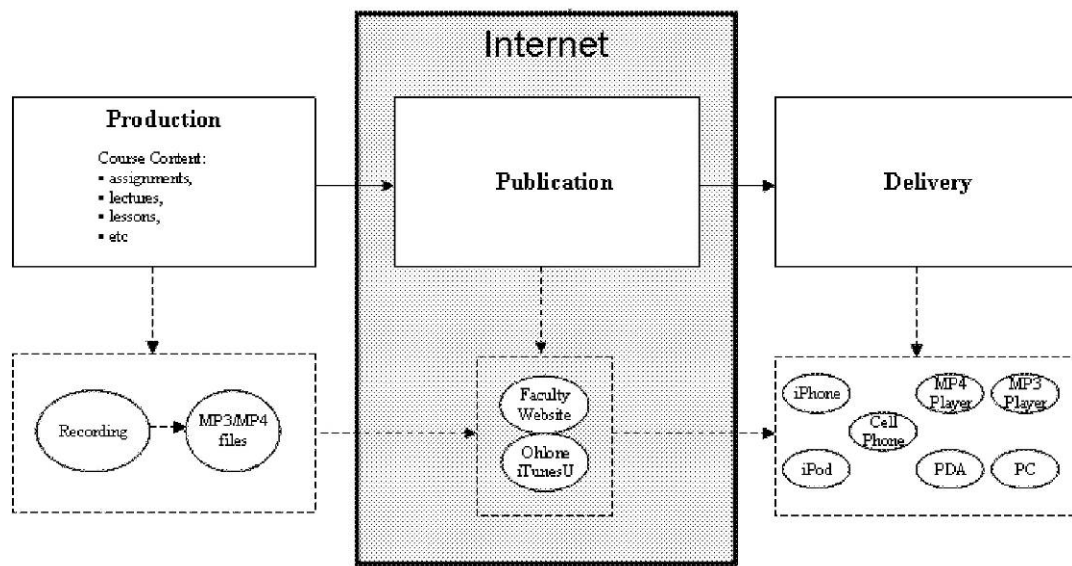


Figure 2. Graphic representation of the production, publication, and delivery of course content through podcasting.

The statistical procedure for two types of educational instructions was the *t* test (Morgan, Leech, Gloeckner, & Barrett, 2007, p. 141). The *t* test was used to test the extent of differences between the two groups of students on the stated measures. For three or more groups (e.g., age groups and number of times listening to the same classroom podcast), a one-way analysis of variance (ANOVA) was used to measure differences (Gay et al., 2006, p. 359; Morgan et al., 2007, p. 157).

There were two independent variables for two groups of students: (a) traditional classroom instruction, and (b) classroom instruction supplemented with classroom podcasts (Table 1). Students were grouped by frequency of classroom podcast usage, class attendance, class levels in college, age groups, gender, and ethnicity. The categorical independent variables of the groupings of students were (a) class attendance (1 = *less than other classes*, 2 = *more than other classes*, 3 = *same as other classes*); (b) class level in college (1 = *1st year*, 2 = *2nd year*, 3 = *3rd year*, 4 = *4th year*, 5 = *5th year or more*); (c) age groups (18 and younger, and 25 and older); (d) gender (1 = *male*, 2 = *female*); and (e) ethnicity (Asian, African American, White, etc.). These variables were used as categorical variables to test differences in measurements. The frequency of classroom podcast usage and students' attitudes toward classroom podcasts were used as ordinal values to rank the students' responses in relation to their test scores (Rugg & Petre, 2007, pp. 182-183). The midterm and final course test scores were used as dependent variables (Appendix A, Tables A1-A3).

Table 1

Participants Grouped by Method of Classroom Instruction

Group	Type of instruction	Designation
1	Traditional classroom instruction	Control group
2	Classroom instruction supplemented with classroom podcast	Treatment group

In statistics there are four levels of measurement: nominal, ordinal, interval, and ratio. Two or more unordered categories are defined as nominal (or ordinal for ordered levels); in interval and ratio scales the data are approximately normal. Two levels or categories are defined as dichotomous variables and three or more levels are defined as categorical variables (Morgan et al., 2007, pp. 36-40).

As the approach for the research design, all students were notified at the first class meeting that classroom podcasts would be provided and posted on Ohlone iTunesU via the instructor's Web page (Appendix B). Detailed instructions on how to use classroom podcasts were provided, along with instructions on how to download and play back the podcasts on cell phones, iPods, MP3 players, or personal computers. The classroom learning materials were recorded as podcasts files and posted on Ohlone iTunesU or the instructor's Web site (Appendix C).

Population

Ohlone College was used as the site for the study. The recruitment method used to select the course instructor was an email to all potential instructors in Ohlone College who used technology in their classroom teaching. The course instructor of the History of the United States class volunteered to participate in the study. All students from the lecture-based course History of the United States (HIST-117A) were invited to participate in the study. Participants received both traditional classroom instruction and classroom instruction supplemented with podcasts. Participation in this study was voluntary and there was no extra grade credit for participation. The sample size of 30 or larger is typically agreed by professional statisticians to be sufficient. According to the central limit theorem, the sampling distribution of the sample mean approximates the normal distribution, regardless of the distribution of the population from which the

samples are drawn, if the sample size is sufficiently large. However, there is no absolute rule that applies in all cases (Boslaugh & Watters, 2008, pp. 137-138). The sample size for the course History of the United States at Ohlone College was 120 students.

Based on Ohlone College's Institutional Research Office (Ohlone College, 2009), the student enrollment for the Fall 2008 semester was 12,842, an increase of 9.1% from fall 2007. A study population size of 72 students from the lecture-based courses History of the United States (HIST-117A) was drawn during the Fall 2009 semester. Within the population the students were grouped by (a) gender, (b) age, (c) year in college, and (d) ethnic groups living in the proximity. In addition, the population included traditional classroom learners and technology-literate learners or experienced users of cell phones, iPods, MP3 players, and personal computers.

The same lecture-based course from the Spring 2009 semester, taught by the same instructors in traditional classroom instruction without podcasting, was used as a control group. Only the midterm and final test scores were selected from the Spring 2009 semester; no personal data were collected. The midterm and final examination test scores from two types of educational instruction (traditional classroom instruction and classroom instruction supplemented with classroom podcasts) were analyzed and used to determine the effectiveness of classroom instruction supplemented with classroom podcasts.

Of the 72 students registered in the lecture-based course History of the United States in the Fall 2009 semester, 85% ($n = 61$) of the students responded to the hand-out survey/questionnaire at the end of the course. The demographic profile of the sample showed that 52.5% of the participating students were female. The age groups ranged from 19-or-younger (50.8%) through 20-24 years (41%) and 25-29 years

(4.9%) to 40-or-older (3.2%). A majority of the respondents (91.8%) was in the 24-or-younger age group. In terms of ethnicity, Whites had the highest participation rate, 42.6%, followed by Asians 19.7%, Hispanics, 11.5% , Multiracial, 11.5%, Filipinos 4.9%, and others 9.8% (Table 2).

Table 2

Distribution of Ethnicity of the Sample

Ethnicity	<i>n</i>	%
Asian	12	19.7
Filipino	3	4.9
Hispanic	7	11.5
White, Non-Hispanic	26	42.6
Multi-racial	7	11.5
Middle Eastern	3	4.9
Hawaiian	2	3.3
Greek	1	1.6

In terms of academic level, second-year college students had the highest participation rate, 59%, followed by 14.8% for third year, 13.1% for first year, 8.2% for fourth year, and 4.9% for students beyond their fifth year (Table 3). In terms of academic major, 19.7% of the participants were in art, 23% in science, 16.4% in business, 6.6% in engineering, 3.3% in history, 8.2% were undecided, and 23% were classified as other (Table 4). In terms of class attendance, 68.9% of the respondents

Table 3

Distribution of Year in College for the Sample

Year in college	<i>n</i>	%
First year	8	13.1
Second year	36	59.0
Third year	9	14.8
Fourth year	5	8.2
Fifth year or more	3	4.9

Table 4

Distribution of the Academic Disciplines of the Sample

Academic major	<i>n</i>	%
Art	12	20.0
Business	10	23.3
Communication	4	16.7
Engineering	4	6.7
Science	14	15.0
Social Science	9	8.3
Undecided	5	6.7
Other	2	3.3

attended class the same number of times as other classes, followed by 26.2% for class attendance more times than other classes and 3.3% for class attendance less than other classes. The results showed that most students attended class lectures regularly and there was no irregularity in students' class attendance for this class compared with other classes (Table 5).

Table 5

Distribution of Class Attendance by the Sample

Class attendance	<i>n</i>	%
Less than other classes	2	3.3
Same as other classes	42	70.0
More than other classes	16	26.7

In terms of accessibility to classroom podcasts, 83.6% of the respondents reported having downloaded classroom podcasts. The majority of respondents (53.3%) were users of personal computers (PC), 16.7% used MP3 players, 8.4% used cell phones, 21.7% used cell phones, MP3 players, or PCs, and 2.0% of students did not respond to the item regarding the listening device (Table 6). The majority of respondents (98.4%) reported ownership of a PC or related PDA device that was used to access Ohlone iTunesU. In terms of familiarity with podcasting prior to the study, 78.7% of the respondents reported that they were familiar with podcasting. The results indicated that the large majority of respondents had no problem in accessing, downloading, and listening to classroom podcasts.

Table 6

Distribution of Listening Devices Reported by the Sample

Listening device	<i>n</i>	%
MP3/iPod player	10	16.7
Personal computer (PC)	32	53.3
MP3 player and PC	13	21.7
Cell phone/PDA, MP3, and PC	4	6.7
Cell phone/PDA and PC	1	1.7

Note. PDA = personal digital assistant.

In terms of students' attitudes toward the use of classroom podcasts, a 5-point Likert-type scale (1 = *strongly disagree*, 2 = *disagree*, 3 = *neutral*, 4 = *agree*, 5 = *strongly agree*) was used. Regarding perceived ease of use of classroom podcasts, 32.8% responded *strongly agree*, 44.3% *agree*, and 23% *neutral*. There was no response for *disagree* or *strongly disagree*. Regarding perceived usefulness of podcasts, 18% responded *strongly agree* and 45.9% responded *agree*. Most respondents chose *strongly agree* or *agree* in response to the item that classroom podcasts increased interest in the subject matter, compared to 26.2% *neutral*, 4.9% *disagree*, and 3.3% *strongly disagree*. The results showed that majority of respondents thought that download and playback of classroom podcasts were easy and that the podcasts were a very useful educational tool (Table 7).

Table 7

Students' Attitudes Toward the Use of Classroom Podcasts

Response	Perceived ease of use (%)	Perceived usefulness (%)
Strongly Agree	32.8	18.0
Agree	44.3	45.9
Neutral	23.0	26.2
Disagree	0	4.9
Strongly Disagree	0	3.3

Data Sources

The data collection instrument was a three-part in-class handout questionnaire used to collect students' responses regarding classroom podcast usage and relevant demographic information (Appendix D). The information captured was used to determine students' learning experiences over the course of an academic semester. All students were invited to respond to the handout questionnaire about their frequency of use of classroom podcasts at the end of the semester. The responses were used to determine the effectiveness of classroom podcasts as teaching and learning tools as measured by midterm and final examination test scores of the students.

The questionnaire included (a) demographics, (b) podcasting, and (c) open-ended questions. The podcasting questions were used to collect data regarding classroom podcast usage. Demographic questions were used to collect general information about participants: gender, classroom attendance, age group, class level in college, and ethnicity. Responses to open-ended questions were used to explain the results. There was nothing in the handout survey/questionnaire that would violate

confidentiality. The questions were not related to participants' psychological well-being, such as family history, welfare, or safety. The questions were purely educational and did not pose a risk for violating confidentiality.

For the demographic questions, *yes/no* questions were used to collect information. For attitude questions, a Likert-type scale of 1-5 was used (1 = *strongly agree*, 2 = *agree*, 3 = *somewhat agree*, 4 = *disagree*, 5 = *strongly disagree*). For the podcasting questions, respondents were directed to check all that applied (Rugg & Petre, 2007, p. 192).

Prior to data collection, permission to conduct the study was obtained from the Office of Student Services at Ohlone College (Appendix E) and the instructors of the target course History of the United States (Appendix F). In order to ensure protection of human subjects, all participants were notified at the beginning of the course that the study was to be conducted over the academic semester and that their information would be anonymous and destroyed after the study. An informed consent letter (Appendix G) was provided to all students prior to their completing the survey/questionnaire at the end of the course.

Not all students participated in the study. Only the submitted survey/questionnaire and valid responses from the students were used in data analysis. The risk assessment was very minimal (Appendix H).

Collected data were coded and stored in SPSS™ version 16 for Microsoft® Windows®. The students' test scores were compared to frequency of classroom podcast usage to determine the effectiveness of classroom podcasts. The students' demographics, class attendance, and class level in college were examined to determine their perceptions of traditional classroom learning and their attitudes toward classroom

podcasts as teaching and learning tools. Statistical analysis software was used to enter and analyze the data.

Limitations of the Study

This study addressed the use of classroom podcasts in one selected lecture-based American history course at Ohlone College. Not all subject areas were represented. All students in this course were invited to participate in the study but not all agreed to participate.

Classroom podcasts provide an information-rich teaching and learning environment that allows educators to distribute classroom learning materials and students to access classroom learning materials on demand. Theoretically, the technique could be replicated to study classroom podcasts at the university level in any subject area. However, podcasting is still a relatively evolving technology, so further study is needed before widespread implementation.

Data Analysis

Data analysis of this study examined the effect of classroom podcasts as supplemental course materials for playback of lectures and other lessons in regard to the average midterm and final examination test scores of the students. Two types of educational instruction (traditional educational instruction and instruction supplemented by listening to classroom podcasts) were examined to determine statistically significant differences in the average midterm and final examination test scores of the students. Descriptive and inferential statistical tests were used in data analysis to compare the groups (Salkind & Rasmussen, 2008, p. 525).

The first step was exploratory data analysis. This step involved descriptive statistics and graphs. This procedure was used as the preliminary step for *t* test and ANOVA (Boslaugh & Watters, 2008, p. 54). Descriptive statistical tests were used to determine central tendency, frequency, distribution, in each category, and variability about the population. These tests were used to test the assumptions of inferential statistical tests. Means and standard deviations were calculated for the two groups. A histogram graph and Q-Q plot were used to conceptualize the data as continuous and to check for normal distribution (Morgan et al., 2007, pp. 36-50).

Statistical inferential tests were used to determine the probability of the differences between two types of educational instruction (traditional classroom instruction and classroom instruction supplemented with classroom podcasts) and test scores obtained from students. A *t* test was used to determine statistical significance between two groups of educational instruction in comparison to their test scores. A *t* test was used to determine statistical significance between two groups of educational instruction, genders, and frequency of classroom podcast usage, as well as student attitudes regarding classroom podcasts. Independent-sample *t* tests and paired-samples *t* tests were used to measure the statistical significance of differences between traditional educational instruction and instruction supplemented by listening to classroom podcasts. For comparison of three or more categorical independent variables, a one-way ANOVA tested the differences between age groups, classroom attendance, and number of hours spent listening to the classroom podcasts.

A null hypothesis was tested for each research question. A null hypothesis posited no difference or no relationship between use of classroom podcasts and test scores (Gay et al., 2006, p. 343; Morgan et al., 2007, p. 92). Statistical significance was determined by the probability level $p < .05$. For an alpha level of less than .05

($p < .05$), the results suggest a difference between the two types of educational instruction (traditional classroom instruction and classroom instruction supplemented with classroom podcasts) and test scores of community college students. For an alpha level of .05 or greater, the results suggest significance and the need for further investigation (Rugg & Petre, 2007, pp. 194-195).

The collected data were coded and analyzed to explore midterm and final examination test scores, frequency of classroom podcast usage, and student attitudes regarding classroom podcasts. Students' test scores were compared to the frequency of classroom podcast usage to measure the effectiveness of classroom podcasts. The students' attitudes toward classroom podcasts were compared to determine the students' perceptions of traditional classroom learning and their attitudes toward classroom podcasts as teaching and learning tools. The results were used to identify, explain, and critique the frequency of classroom podcast usage.

Summary

The methodological approach for this study is outlined in this chapter. The study was designed to determine the effectiveness of classroom podcasts (as measured by reported frequency of use) in relation to the students' test scores. The study used a quantitative experimental design. A handout survey/questionnaire was used to collect data. The collected data were used to compare the effectiveness of classroom podcast usage and students' test scores.

Chapter 4

RESULTS

Chapter 4 presents the results of a quantitative data analysis for a case study of classroom podcast in Ohlone College. Traditional classroom instruction and classroom instruction supplemented with classroom podcasts were examined. A quasi-experimental design was used to study the two types of educational . This study investigated the use of podcasting in reinforcing learning outcomes for community college students. A handout survey/questionnaire was used to collect students' evaluative responses about their learning experiences with classroom podcasts. The survey was conducted at the end of the Fall 2009 semester. The survey data were analyzed using SPSS to determine the effectiveness of classroom podcast usage on learning outcomes. Results were examined to determine the students' learning experiences over the course of one academic semester and their perceptions of classroom podcasts.

The thesis of this study was that students who listen to classroom podcasts are more likely to achieve higher test scores than students who do not listen to classroom podcasts. First, the effectiveness of podcasting on summative evaluation of learning outcomes was measured by the students' achievement on the course midterm and final examinations. Learning outcomes were based on the test scores for the whole group. Higher average test scores of the whole group indicated better learning outcomes. According to Salkind and Rasmussen (2008), test scores are commonly used to assess how well learners have mastered learning concepts. Second, the data reported the

students' perceptions of classroom podcasts as determined by respondents' frequency of use, age group, class attendance ethnic, and gender. The study reported results of the effects of mobile and portable media devices (iPods and MP3/MP4 players) and personal computers, and the impact of these devices on student learning outcomes via classroom instruction with supplemental audio course materials. The prediction was that, when students feel comfortable with these technologies, they will be more likely to adopt podcasting as teaching and learning method. Each of the hypotheses was analyzed to determine the effectiveness of podcasting as a teaching and learning method.

The research employed a quasi-experimental design. Two intact groups (students in the course History of the United States during Fall 2009 and Spring 2009 semesters) were examined to determine the effectiveness of classroom podcast usage on the students' test scores. The treatment group of participants received classroom instruction supplemented with classroom podcasts. The comparison group of participants received classroom instruction without classroom podcasts. The midterm and final examination scores of the two groups were compared to determine the effectiveness of the treatment. The same instructor taught both courses with the same learning objectives and the same grading criteria. Midterm and final examination test scores of the two groups were examined. The means of midterm and final examination test scores were calculated to determine the significant differences between the two groups (Salkind & Rasmussen, 2008, pp. 177-178).

A quantitative data analysis was based on the students' responses to the hand-out survey/questionnaire at the end of the course. The questionnaires were organized in three parts: (a) demographics, (b) podcasting, and (c) open-ended questions. Demographic questions were used to collect general information about participants: age

group, class attendance, class level in college, gender, and ethnicity. Podcasting questions were used to collect data regarding the frequency of use of classroom podcasts and the students' attitudes toward classroom podcasts. Open-ended questions were used to provide explanation of the results. Inferential statistics were used to test each of the research questions and hypotheses. The statistical analysis software tool SPSS was used to code, store, and analyze the data.

The results focused on the differences between two types of educational instruction (traditional classroom instruction and classroom instruction supplemented with classroom podcasts). The independent-samples t test was used to determine statistically significant differences between the two types of educational instruction as measured by the students' test scores (midterm and final examinations). The paired-samples t test was also used to compare differences between the midterm and final examination test scores of the two groups. In addition, the paired-samples t test was used to determine the reliability of the midterm and final examinations (test and retest). The tests were analyzed using SPSS (Morgan et al., 2007, p. 152).

Within the treatment group, the independent-samples t test was used to compare the differences between genders with regard to the number of hours per week students listened to classroom podcasts. For three or more categorical independent variables (age groups and class attendance), one-way ANOVA was used to analyze the data. Both the independent-samples t test and one-way ANOVA were based on the .05 level of significance (Salkind & Rasmussen, 2008, pp. 832-837).

Results for Research Question 1

Research question 1 asked, Are there significant differences between two types of educational instruction (traditional classroom instruction and classroom instruction

supplemented with classroom podcasts) in regard to the average midterm and final examination scores of students?

Two types of educational instruction (1 = *traditional classroom instruction* and 2 = *classroom instruction supplemented with classroom podcasts*) were examined. The treatment group received classroom instruction supplemented with classroom podcasts and the comparison group received traditional classroom instruction. The test scores of the two groups were compared to determine the effectiveness of the treatment. Each of the test scores from the two groups was a discrete number ranging from low (0) to high (100) and independent from other scores. The distribution of the midterm and final examination test scores showed that the two sampling distributions were approximately straight lines and normally distributed (Figure 3). The assumptions for use of the independent-samples t test and paired-samples t test were met (Boslaugh & Watters, 2008, pp. 118-119; Morgan et al., 2007, pp. 149-150).

An independent-samples t test was calculated for the final examination test scores of the treatment group ($n = 72$) and comparison group ($n = 70$). Table 8 shows the means and standard deviations for the treatment group ($M = 77.78$, $SD = 11.44$). The Levene test to check the assumption was significant ($p < .01$) and the assumption of equal variances was violated. The results indicated no statistically significant differences between the two groups ($t = 1.32$, $df = 126.77$, $p < .19$). Table 9 shows the results of the midterm examination scores.

A repeated independent-samples t test was calculated for repeated measures (pretest and posttest) on the treatment group (posttest $n = 72$) and comparison group (posttest $n = 70$). The Levene test to check the assumption was significant ($p < .006$) and the assumption of equal variances was violated. The results indicated no statistically significant differences between the two groups ($t = .30$, $df = 129.74$,

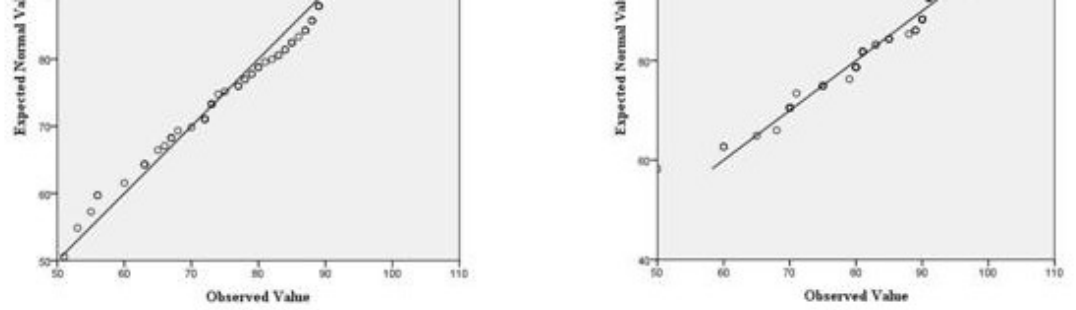


Figure 3. Normal Q-Q Plot of midterm test scores and final examination test scores of the treatment and comparison groups.

$p < .765$). Table 10 shows the means of the treatment and comparison groups. Table 11 shows the results of the final examination.

Table 9

Results of Independent-Samples Two-Tailed t Test for Comparison of the Two Types of Educational Instruction (Traditional Classroom Instruction and Classroom Instruction Supplemented With Classroom Podcasts) on Midterm and Final Examination Test Scores

Test	Treatment group		Comparison group	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Midterm	77.80	11.44	74.76	15.51
Final examination	80.29	9.19	79.76	11.91

Table 9

Descriptive Statistics for Midterm and Final Examination Test Scores

Test	<i>df</i>	<i>t</i>	Significance (2-tailed)
Midterm			
Final examination			

A paired-samples *t* test was calculated for repeated measures (midterm and final examinations) on the treatment group. Analysis of the 72 matched pairs showed the results of the treatment (Table 10). The means of the midterm test scores and the final examination test scores were significantly different. Mean scores for midterm ($M = 77.78$, $SD = 11.44$) were lower than final examination test scores ($M = 80.29$, $SD = 9.19$). The difference was statistically significant ($N = 72$; $df = 71$; $p < .038$). Correlated paired-samples *t* tests indicated a moderate relationship between the two test scores ($r = .538$, $p < .001$).

Compared to the 70 matched pairs of the comparison group, the means for the midterm and final examinations were significantly different, consistent with the paired-samples t test for the treatment group. The mean test scores for the midterm ($M = 74.76$, $SD = 15.51$) were lower than the mean test scores for the final examination ($M = 79.90$, $SD = 11.97$). The difference was statistically significant ($N = 70$, $df = 69$, $p < .002$). Table 10 reports the results for the comparison group. Correlated paired-samples t tests indicated a moderate relationship between midterm and final examination scores ($r = .598$, $p < .001$).

Table 1

Results of Paired Samples Two-Tailed t Test for the Comparison of Midterm Examination

2.	.0

Table 10

Results of Paired Samples Two-Tailed t Test Comparing Midterm and Final Examination Test Scores of Treatment and Comparison Groups

	N	Correlation	Significance
The treatment group	72	.538	.001
The comparison group	70	.587	.001

Results for Research Question 2

Research question 2 asked, *Is there a significant difference between the number of times students listen to classroom podcasts before and after the midterm?*

The number of times students listen to classroom podcasts before and after the midterm was measured using a 5-point Likert-type scale (0 = *not at all*, 1 = *once*, 2 = *twice*, 3 = *three times*, and 4 = *four or more times*) for the treatment group. Each of the five data points was a discrete number and independent from other points. The two sampling distributions for the number of times students listened to classroom podcasts before and after the midterm examination showed that the two distributions were approximately straight lines and normally distributed (Figure 4). The assumptions for use of the paired-samples t test were met (Morgan et al., 2007, pp. 149-150).

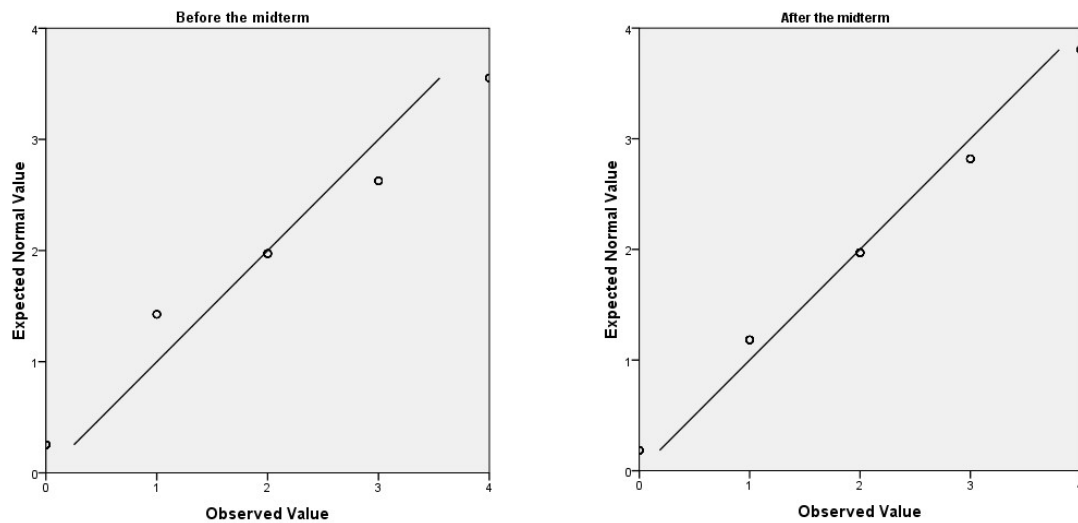


Figure 4. Normal Q-Q plot of the number of times students listened to classroom podcasts before and after the midterm examination.

Analysis of the 60 matched pairs by a paired-samples t test showed a significant difference between the number of times students listened to classroom podcasts before the midterm ($M = 1.33$, $SD = 1.49$) and after the midterm ($M = 1.75$, $SD = 1.32$). The difference was statistically significant ($N = 60$, $df = 59$, $p < .012$). The null hypothesis (H_0) of the two paired-samples t tests was rejected and the alternative hypothesis (H_a) was accepted. Table 11 reports the results for the treatment group. Table 12 shows that correlated paired-samples t test indicated a moderate relationship between the two variables ($r = .618$, $p < .01$).

Table 11

Results of Paired Samples Two-Tailed t Test for the Comparison of the Number of Times Students Listened to Podcasts Before and After the Midterm Examination

	N	t	df	Significance (2-tailed)
Before and after the midterm	60	2.60	59	.012

Table 12

Results of Paired Samples Correlation for Comparison Between of the Number of Times Students Listened to Podcasts Before and After the Midterm Examination

	N	Correlation	Significance
Before and after the midterm	60	.618	.001

Results for Research Question 3

Research question 3 asked, *Is there a significant difference between class attendance (less than other classes, same as other classes, and more than other classes) with regard to the number of hours students listen to classroom podcasts per week?*

The distribution of the number of hours per week students listened to classroom podcasts had 4 data points (0 = *not at all*, 1 = *1 hour or less*, 2 = *2 hours*, and 3 = *3 hours or more*); each was a discrete number and independent from the other numbers. The number of hours per week students listened to classroom podcasts was classified as an ordinary variable (Morgan et al., 2007, pp. 38-40). With only 4 data points, a histogram with normal curve showed that the underlying graph was approximately normal (Figure 5). This variable was classified as ordinary variable (Gay et al., 2006, p. 124). Class attendance was classified as a categorical independent variable with three levels (1 = *less than other classes*, 2 = *same as other classes*, 3 = *more than other classes*).

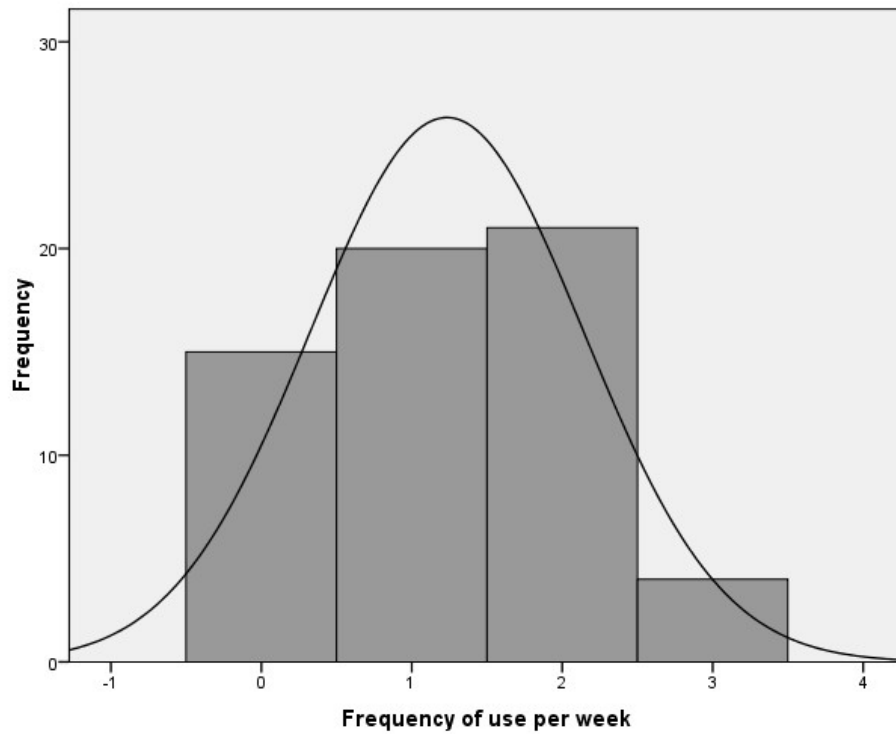


Figure 5. Histogram with normal curve depicting the number of times per week students listened to classroom podcasts.

A one-way ANOVA was used to determine statistically significant difference between class attendance and number of hours the students listened to classroom podcasts. Table 13 shows the descriptive statistics for the three levels class attendance. Table 13 shows that class attendance less than other classes ($n = 2$, $M = 1$, $SD = 0$) was

Table 13

Descriptive Statistics for Class Attendance and the Number of Hours Per Week Students Listened to Classroom Podcasts

Class attendance	<i>n</i>	<i>M</i>	<i>SD</i>
Less than other classes	2	1.00	0.000
Same as other classes	42	1.21	0.925
More than other classes	16	1.31	0.946
Total	60	1.23	0.909

significantly lower than class attendance the same as other classes ($n = 42$, $M = 1.21$, $SD = .925$) or class attendance more than other classes ($n = 16$, $M = 1.31$, $SD = .946$).

Table 14 shows that results of the Levene test to check the assumption were not significant ($p < .071$), which suggests that the two variances were approximately equal.

The nonparametric Kruskal-Wallis test was used to test the reliability of the measure because the two variances were approaching significance and the dependent variable had 4 data points (Morgan et al., 2007, pp. 166-168).

Table 14

Results of Test of Homogenous Variance for Class Attendance and the Number of Hours Per Week Students Listened to Classroom Podcasts

	Significance
Frequency of use	.

A one-way ANOVA found no statistically significant between class attendance and the number of hours per week students listened to classroom podcasts (Table 15). The difference was not statistically significant, $F(2, 57) = .132, p < .877$. The null hypothesis (H_a) was accepted. The test observed only respondents who participated in the handout survey/questionnaire and did not account for students who did not participate.

Table 15

Results of One-Way Analysis of Variance for Comparison of Class Attendance and the Number of Hours Per Week Students Listened to Classroom Podcasts

Group	<i>df</i>	<i>F</i>	Significance (2-tailed)
Frequency of use	2	1.32	.877

Results for Research Question 4

Research question 4 asked, *Is there a significant difference between age groups (19 or younger, 20-24, and 25 or older) with regard to the number of hours students listen to classroom podcasts per week?*

The distribution of the number of hours per week students listened to classroom podcasts had 4 data points: (0 = *not at all*, 1 = *1 hour or less*, 2 = *2 hours*, and 3 = *3 hours or more*). Each of the four data points was a discrete number and independent from other data points. The number of hours per week students listened to classroom podcasts was classified as ordinary variable. With only 4 data points, a histogram with normal curve showed that the underlying graph was approximately

normal (Figure 5). Respondents were grouped by age groups (1 = 19 or younger, 2 = 20-24, 3 = 25 or older).

A one-way ANOVA was used to determine statistical significance of the differences among age groups according to the number of hours per week they listened to classroom podcasts. Table 18 shows that the mean for the 19-or-younger age group ($n = 30$, $M = 1.17$, $SD = .874$) was slightly higher than the means for the 20-24 age group ($n = 25$, $M = 1.16$, $SD = .943$) and the 25-or-older age group ($n = 5$, $M = 2$, $SD = .707$). Table 19 shows that the Levene test to check the assumption was not significant ($p < .322$). The two variances were approximately equal on the dependent variable.

Table 18

Descriptive Statistics by Age Group for the Number of Hours Per Week Students Listened to Classroom Podcasts

Age group	<i>n</i>	<i>M</i>	<i>SD</i>
19 years or younger	30	1.17	.874
20 to 24 years	25	1.16	.943
25 years or older	5	2	.707
Total	60	1.23	.909

A one-way ANOVA found no statistically significant difference by age group according to the number of hours per week students listened to classroom podcasts, $F(2,57) = 2.01$, $p < .144$. The null hypothesis (H_0) was accepted. The test observed

Table 19

Results of Test of Homogenous Variance of Age Groups and the Number of Hours Per Week Students Listened to Classroom Podcasts

	Levene statistic	<i>df1</i>	<i>df2</i>	Significance
Frequency of use	1.158	2	57	.322

only respondents who participated in the handout survey/questionnaire and did not account for students who did not participate. Table 20 reports the results of the one-way ANOVA.

Table 20

Results of One-Way Analysis of Variance for Comparison of Age Groups and Number of Hours Per Week Students Listened to Classroom Podcasts

	<i>df</i>	<i>F</i>	Significance (2-tailed)
Frequency of use	2	2.01	.144

While there was no significant difference between the two variables, cross-tabulation of age groups and number of hours per week students listened to classroom podcasts indicated that, overall, the 19-or-younger age group and the 20-to-24 age group were higher than the 25-or-older age group. However, the results also indicated that the 25-or-younger age group was more likely than were the 24-or-younger age

group to listen to classroom podcasts (Table 21). There was a weak association between the two variables ($\eta = .103$).

Results for Research Question 5

Research question 5 asked, *Is there a significant difference between genders (male and female) with regard to the number of hours students listen to classroom podcasts per week?*

The number of hours per week students listened to classroom podcasts had 4 data points (0 = *not at all*, 1 = *1 hour or less*, 2 = *2 hours*, and 3 = *3 hours or more*) and was designated an ordinary variable, each with a discrete number independent of other data points. Gender had two nominal categories (1 = *female* and 2 = *male*).

An independent-samples t test was used to compare number of hours per week students listened to classroom podcasts by gender. With only 4 data points, a histogram with normal curve showed that the underlying graph was approximately normal, especially near the center (Figure 5).

An independent-samples t test found a statistically significant difference between male and female students on number of hours per listening to classroom podcasts. Table 22 shows that the mean for females students ($n = 32$, $M = 1.5$, $SD = .916$) was higher than the mean for male students ($n = 28$, $M = .93$, $SD = .813$; Table 23). The difference was statistically significant ($df = 60$, $t = 2.54$, $p < .014$). The null hypothesis (H_0) was rejected and the alternative hypothesis (H_a) was accepted.

Table 14

Results of Levene's Test for Equality of Variances for Class Attendance and the Number of Hours Per Week Students Listened to Classroom Podcasts

	<i>F</i>	Significance
Frequency of use	.5	1

Table 22

Descriptive Statistics for Number of Hours Per Week Students Listened to Classroom Podcasts by Gender

Gender	<i>n</i>	<i>M</i>	<i>SD</i>
Female	32	1.50	0.916
Male	28	0.93	0.813

Table 23

Results of Independent-Samples Two-Tailed t Test for Number of Hours Per Week Students Listened to Classroom Podcasts by Gender

	<i>t</i>	<i>df</i>	Significance (2-tailed)
Frequency of use	2.54	58	.014

Additional Findings

Some factors not addressed by the research questions were found to be important complements to the findings related to the research questions hypotheses. These six additional findings are reported in this section.

1. An open-ended (nonempirical) question (the last question in the handout survey/questionnaire) was posed: “*How could podcasts be improved to better meet your needs?*” Of the 61 respondents, 82% ($n = 50$) answered this question. Written responses were assigned to four categories: (a) quality of classroom podcasts, (b) podcasting as a teaching and learning tool, (c) learning preferences, and (d) recommendations for improvement.

Regarding the quality of classroom podcasts, 15 respondents stated that the podcasts were good in their current format. Comments included the following: (a) Classroom podcasts should be made available to the world, (b) “I did well on the midterm so the podcasts were not needed,” and (c) “a huge technological step in the right direction for students.” Regarding podcasting as a teaching and learning tool, respondents stated that podcasts were helpful for reviewing class lectures ($n = 13$) and for reviewing interaction between teacher and students ($n = 1$). Regarding learning preferences, respondents preferred classroom lectures ($n = 10$), audio/video podcasts ($n = 6$), and MP3 or compatible audio file formats in addition to iTunes ($n = 3$). Recommendations for improving the use of podcasts included (a) arrange podcasts in chronological order ($n = 4$), (b) provide better access to the podcasts other than iTunesU ($n = 1$), (c) provide video podcasts for visual learners ($n = 3$), (d) provide better sound quality to hear the interaction between teacher and students ($n = 3$), (e) edit to remove extraneous material from the podcasts ($n = 3$), and (f) expand podcasting by offering services to other classes ($n = 2$). These responses were highly variable and difficult to code but it is presumed that the responses reflected students’ true opinions at the time of the survey (Table 24).

Table 24

Distribution of Responses to the Open-Ended Question Asking for Recommendations for Improving Podcast Services (N = 50)

Category	<i>f</i>	%
Quality of classroom podcasts		
Good the way it is	15	30
Classroom podcasts should be made available to the world	1	2
Did well on the midterm so don't need the podcast	1	2
Technological step in the right direction for students	1	2
Podcasting as a teaching and learning tool		
Helpful for reviewing class lectures	13	26
Helpful for reviewing interaction between teacher and students	1	2
Learning preferences		
Preferred classroom lectures	10	20
Preferred audio/video podcast	6	12
Preferred MP3 or compatible audio file format	3	6
Recommendations for improvement		
Arrange the podcasts in chronological order	4	8
Provide video podcast for visual learners	3	6
Provide better sound quality	3	6
Remove extraneous material from the podcasts	3	6
Expand use of the podcasts to other classes	2	4

2. Is there a significant difference by level of familiarity with podcasting with regard to the number of hours per week students listen to classroom podcasts? An independent-samples t test found no statistically significant relationship between level of familiarity with podcasting and number of hours per week the students listened to classroom podcasts. Table 25 shows that the mean of yes responses ($n = 32$, $M = 1.17$,

Table 25

Descriptive Statistics for Comparison of Number of Hours Per Week Students Listened to Podcasts by Level of Familiarity With Podcasting

Familiarity with podcasting	<i>n</i>	<i>M</i>	<i>SD</i>
Female	48	1.17	1.31
Male	12	1.50	2.61

SD = 1.31) was significantly lower than the mean of no responses, (n = 12, M = 1.50, SD = 2.61; Table 26). The difference was no statistically significant difference (df = 58, t = -1.14, p < .259).

Table 26

Results of Independent-Samples Two-Tailed t Test for Number of Hours Per Week Students Listened to Classroom Podcasts by Level of Familiarity with Podcasting

	<i>t</i>	<i>df</i>	Significance (2-tailed)
Frequency of use	-1.14	58	.259

3. *Is there an association between ethnicity and the number of hours per week students listened to classroom podcasts?* Eta was used to investigate the strength of the association between ethnicity and the number of hours per week students listened to classroom podcasts. Results showed a weak relationship between the two variables

($\eta = .196$). However, overall, cross-tabulation showed that Hispanics were lower than all other ethnic groups in the frequency of listening to podcasts (Table 27).

Table 18

Descriptive Statistics by Age Group for the Number of Hours Per Week Students Listened to Classroom Podcasts

Age group	<i>n</i>	<i>M</i>	<i>SD</i>
Asian		1.	.
Filipino		1.	.
Hispanic	25	1.32	.945
Multiracial	7	1	.816
Others	6	1.67	.577

Table 27

Cross-Tabulation of Ethnicity Versus Number of Hours Per Week Students Listened to Classroom Podcasts

Ethnic group		Zero	1 hour or less	2 hours	3 hours or more	Total
Asian	Count	3	4	4	1	12
	%	25.0	33.3	33.3	8.3	100.0
Filipino	Count	1	0	2	0	3
	%	33.3	0.0	66.7	0.0	100.0
Hispanic	Count	3	2	2	0	7
	%	42.9	28.6	28.6	0.0	100.0

White, Non-Hispanic					
Count	5	10	7	3	25
%	20.0	40.0	28.0	12.0	100.0
Multiracial					
Count	2	3	2	0	7
%	28.6	42.9	28.6	0.0	100.0
Others					
Count	1	1	4	0	6
%	16.7	16.7	66.7	0	100.0
Total					
Count	15	20	21	4	60
%	25.0	33.3	35.0	6.7	100.0

4. *Is there an association between years in college and number of hours per week students listened to classroom podcasts?* Eta was used to investigate the strength of the association between year in college and number of hours per week students listened to classroom podcasts. The results showed no relationship between the two variables ($\eta = .081$). However, cross-tabulation showed that students in the second and fourth years in college reported a higher frequency of use than students in all other years (Table 28).

Table 28

Cross-Tabulation of Year in College Versus Number of Hours Per Week Students Listened to Classroom Podcasts

Year in college		Zero	1 hour or less	2 hours	3 hours or more	Total
First	Count	3	2	3	0	8
	%	37.5	25.0	37.5	0.0	100.0

Second	Count	6	12	14	3	35
	%	17.1	34.3	40.0	8.6	100.0
Third	Count	4	3	1	1	9
	%	44.4	33.3	11.1	11.1	100.0
Fourth	Count	1	2	2	0	5
	%	20.0	40.0	40.0	0.0	100.0
Fifth or more	Count	1	1	1	0	3
	%	33.3	33.3	33.3	0.0	100.0
Total	Count	15	20	21	4	60
	%	25.0	33.3	35.0	6.7	100.0

5. *What was the percentage of the number of times students listened to the same podcast?* Descriptive statistics showed that the highest rate was 45.9% for a one-time listen to the same podcast. However, 16.4% of the respondents reported listening to the same podcast 4 or more times, 11.5% for 2 times, and 6.6% for 3-times; 11% reported not listening to the classroom podcasts at all (Figure 6).

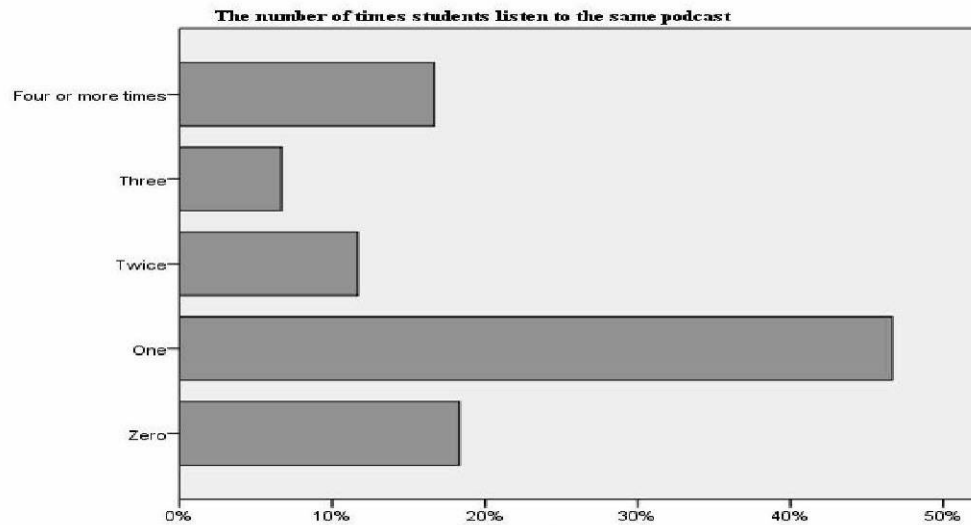


Figure 6. Number of times students reported listening to the same podcast.

6. *What is the percentage distribution of final course grades for classroom instruction supplemented with classroom podcasts?*

For the 72 students in the classroom instruction supplemented with classroom podcasts the distribution of final course grades was as follows: 30.6% received an A, 38.9% received a B, 23.6% received a C, 5.6% received a D, and 1.4% received an F. Results showed that most students received a grade of B (Table 29).

For the 70 students receiving classroom instruction with no classroom podcast, the distribution of final course grades was as follows: 34.3% received an A, 25.7%

Table 29

Distribution of Final Course Grades for Classroom Instruction Supplemented With Classroom Podcasts

Final course grade	<i>n</i>	%
A	22	30.6
B	28	38.9
C	17	23.6
D	4	5.6
F	1	1.4

received a B, 30.0% received a C, 5.7% received a D, and 4.3% received an F. Results showed that the majority of students received a grade of A (Table 30).

Table 30

Distribution of Final Course Grades for Traditional Classroom Instruction With No Classroom Podcasts

Final course grade	<i>n</i>	%
A	24	34.3
B	18	25.7
C	21	30.0
D	4	5.7
F	3	4.3

Summary

The results of this study showed no significant difference in the average midterm and final examination test scores between traditional classroom instruction (the comparison group) and classroom instruction supplemented with classroom podcasts (the treatment group). However, the average test scores of the treatment group were slightly higher than those of the comparison group.

Of the 60 respondents 81.7% reported listening to the podcasts at least once during the semester. Before the midterm, 53.3% of respondents reported that they had listened to the podcasts at least once. After the midterm 76.7% of the respondents reported having listened to the podcasts at least once and 18.3% reported not having listened to the podcasts at all. Results related to research question 2 showed a statistically significant difference between the number of times students listened to classroom podcasts before and after the midterm examination.

The results of data analysis showed that the majority of the respondents reported having listened to the podcasts at least once during the semester. Before the midterm examination the majority of respondents reported that they had listened to the podcasts at least once. After the midterm examination a much larger majority of respondents reported having listened to the podcasts at least once. Nearly 20% of the respondents reported not having listened to the podcasts at all. Results related to research question 2 showed a statistically significant difference between the number of times students listened to classroom podcasts before and after the midterm examination. Results related to research question 5 showed significant gender differences in number of hours per week students listened to classroom podcasts: Female students were more likely to listen to class podcasts.

Overall, the results of data analysis showed no statistically significant differences between classroom instruction and classroom instruction supplemented with classroom podcast. However, there were differences in midterm and final examination test scores: The final examination test scores were significantly higher than the midterm test scores. Gender and number of times students listened to classroom podcasts before the midterm and after the midterm also differed significantly. Based on descriptive statistics for the groups, the large majority of respondents reported having taken advantage of classroom podcasts and a relatively small number of the respondents reported not having used the podcasts. Analysis by age group, ethnic group, and class attendance showed no statistically significant relationship to number of hours per week students listened to classroom podcasts. The results showed that a large majority of the students had a very positive view of classroom podcasts.

Chapter 5

CONCLUSIONS, DISCUSSION, AND RECOMMENDATIONS FOR FUTURE RESEARCH

As with many of the today's news media and entertainment outlets, the purpose of podcasting is simultaneously to distribute information via audio or video podcast and retrieve the information when the podcast episode is over. Like the news media and entertainment outlets, podcasting has expanded from personal entertainment usage to educational purposes, by which training course content may be accessed via personal computers, cell phones, and portable media players. Podcasting allows faculty to create, publish, and post course content on Web sites (Barlow, 2008; Mikat et al., 2007). Students can access and download the podcasts either from faculty Web sites or from Ohlone's iTunesU, to play back on their personal entertainment devices.

With podcasting of course content, students have additional time to review classroom lectures, lessons, and assignments. This method offers students an alternative reinforcement for learning key concepts in the course. Based on theories of humanism and behaviorism, reinforcement is imperative to student learning outcomes by helping students to achieve competency in what is being learned (Salkind & Rasmussen, 2008). These concepts are essentially cognitive style preferences (sensory preferences, environmental preferences, instructional preferences, social and emotional learning aspects, and thinking preferences) that help learners to think.

According to Barlow (2008) and Fryer (2008), students are likely to be more engaged in subject matter when they are interested in learning content with which they are familiar. With the learning content in the form of entertainment and leisure, learning is fun and exciting.

Statement of the Problem

Like the Internet, podcasting has grown at a remarkable rate. The Pew Internet and American Life Project (2005) reported that almost 9 in 10 teens were Internet users and approximately 11 million teens went online daily. About 87% of teens in the United States used the Internet—about 21 million youth, up from about 17 million in late 2000. The numbers of teenage using the Internet and related technology have increased in a variety of arenas, including higher education. A report by the Pew Internet and American Life Project (2009b) showed that mobile phone use had climbed 63% in the 12- to 17-year-old group from fall 2006 to 71% in early 2008. They also reported that 74% of teens owned an iPod or MP3 player and 60% of teens owned a personal computer.

With rapid growth in the general population and in higher education, podcasting has fundamentally changed classroom teaching and the learning environment. Many academic institutions, including Duke University, Harvard University, MIT, and Stanford University, have already offered course materials via audio/video podcasts (Chan et al., 2006; Fryer, 2008). Read (2005) reported that in 2004 Duke University distributed iPods to all first-year students as part of an initiative to encourage their use in education. After evaluating the program, the university revolutionized the adoption of the MP3 media players as a course content dissemination tool. Not only is podcasting offered at leading universities; Ohlone College, the site for this study,

launched its own Ohlone's iTunesU, aimed at providing capability for students to access and download audio and video podcasts to their computers or iPods directly from iTunes (Ohlone College, 2006).

Although learning through podcasting methods offers new flexible learning outside the classroom, it also presents challenges to traditional classroom learning. One of the major challenges facing classroom learning is how to engage the diverse learning styles of student, especially community college students (Lyles et al., 2007). With the flexibility of classroom podcast learning, students choose their preferred learning styles to succeed in course work. However, the benefits of incorporating classroom podcasts into traditional classroom settings remain unclear. This study examined classroom podcast usage in relation to the students' learning outcomes as measured by midterm and final examination test scores.

Review of Methodology

A quasi-experimental design was used to determine the effect of classroom podcasts in regard to the average midterm and final examination test scores of students at Ohlone College. Two types of educational instruction (classroom instruction alone and classroom instruction supplemented with podcasts) were compared to determine the effects of the use of classroom podcasts. Traditional classroom instruction (attending lectures, reading the textbook, and taking notes) was used by the comparison group. Quantitative data analysis was used to determine the effectiveness of classroom podcasts in relation to the students' test scores.

Two intact groups (students enrolled in the course History of the United States in Fall 2009 and Spring 2009 semesters) were examined as groups (not individually) to determine the effectiveness of the classroom podcast usage on the students' test

scores. The treatment group received classroom instruction supplemented with classroom podcasts; the comparison group received traditional classroom instruction without classroom podcasts. The midterm and final examination test scores of the two groups were compared to determine the effectiveness of the use of podcasts. The same instructor taught both courses with the same learning objects and the same grading criteria.

A study sample of 91 students from the lecture-based courses History of the United States (HIST-117A) was drawn during the Fall 2009 semester at Ohlone College. A podcast of the classroom lecture for each of the meetings was published on Ohlone's iTunesU and WebCT. All enrolled students were invited to participate in the study. The student population included male and female students divided by age into groups of 19 years or younger, 20 to 24 years, 25 to 29 years, 30 to 34 years, 35 to 39 years, 40 to 49 years, and 50 years or older. The students' demographic profile included year in college (first, second, third, fourth, fifth, or more years) and diverse ethnic groups.

The data collection instrument was a three-part questionnaire that was used to collect information about the students' learning experiences over the course of the Fall 2009 semester. The students' responses to the handout server/questionnaire were used to determine classroom podcast usage in relation to the learning outcomes as measured by the midterm and final examination test scores. The questionnaire was organized into three parts: (a) demographics, (b) use of podcasting, and (c) open-ended questions. Demographic questions collected general information about the participants: gender, age group, class attendance, class level, and ethnicity. Podcasting questions collected the frequency of use of classroom podcasts and the attitude toward podcasting. Open-ended questions collected additional information. The participants were

asked to respond to the handout survey/questionnaire about classroom podcast usage and their learning experiences.

The collected data were coded and analyzed to explore midterm and final examination test scores, frequency of classroom podcast usage, and student attitudes regarding classroom podcasts. The students' test scores were compared to frequency of classroom podcast usage to determine the presumed effectiveness of classroom podcasts on test scores. Inferential statistics were used to measure differences between and within groups of students (traditional educational instruction and classroom instruction supplemented with classroom podcasts). The statistical analysis software tool SPSS[®] (Norusis, Inc.) was used to analyze the data.

Conclusion and Finding for Research Question 1

Research question 1 asked, *Are there differences between two types of educational instruction (traditional classroom instruction and classroom instruction supplemented with classroom podcasts) in regard to the average midterm and final examination scores of students?*

The study found no statistically significant difference between the average and midterm final examination test scores of students who participated in classroom instruction supplemented with classroom podcasts (the treatment group) and students who participated in classroom instruction without classroom podcast (the comparison group). However, the midterm and final examination test scores of the treatment group were higher than those of the comparison group. Within the treatment group, final examination test scores were significantly higher than midterm examination test scores.

Although there was not much difference between the midterm and final examination test scores of the two groups, there was evidence to suggest that students who listened to classroom podcasts were more likely to achieve higher test scores than students who did not listen to classroom podcasts. The results suggest that students who listened to classroom podcasts for the midterm examination were likely to listen to classroom podcasts for the final examination. Students who scored high on the midterm examination were also likely to score high on the final examination. Students who had higher test scores on the midterm and final examinations were less likely to take advantage of classroom podcasts, especially those students who preferred class lectures.

Overall, this study found that students were likely to be motivated by self-directed learning and were likely to take advantage of any available tools to learn. Although some students found classroom podcasts helpful for understanding the subject matter, podcasting was clearly not attractive or helpful for every student. While there are many factors affecting student learning, the main factor seems to be the student's desired to receive higher course grades. Nonetheless, it was concluded that the majority of the students had positive views of podcasting as a teaching and learning tool.

Conclusion and Finding for Research Question 2

Research question 2 asked, *Is there a significant difference between the number of times students listened to classroom podcasts before and after the midterm?*

The data showed that the number of times students listened to classroom podcasts before and after the midterm examination differed significantly. The comparison of the average number of times students listened to classroom podcasts before and

after the midterm examination showed that more students listened to classroom podcasts after the midterm examination than before the examination. The results showed a positive relationship between the number of times students listened to classroom podcasts before and after the midterm examination.

Based on these results, it was concluded that students who engaged in learning with classroom podcasts were likely to be motivated by course grades. The results suggest that students who listened to classroom podcasts before the midterm examination were more likely to listen to classroom podcasts after the midterm examination. Similarly, students who chose not to listen to classroom podcasts before the midterm examination were likely to listen to classroom podcasts before the final examination.

Conclusion and Finding for Research Question 3

Research question 3 asked, *Is there a significant difference between class attendance with regard to the number of hours per week students listened to classroom podcasts?*

The data showed found no statistically significant relationship between classroom attendance and the number of hours per week students listened to classroom podcasts. The comparison of the two groups revealed that students listened to classroom podcasts regardless of their level of class attendance.

Based on the results, it was concluded that the large majority of students who attended class lectures regularly listened to classroom podcasts. Likewise, students who chose not to take advantage of classroom podcasts attended class normally. The results suggest that class attendance was not a factor in usage of classroom podcasts. Although these findings were considered to be extraneous, they provide support for

the conclusions regarding research question 1 and research question 2 that class attendance had no impact on the use of classroom podcasts.

Conclusion and Finding for Research Question 4

Research question 4 asked, *Is there a significant difference between the age groups on the number of hours per week students listened to classroom podcasts?*

The data showed no statistically significant difference by age regarding the number of hours per week students listened to classroom podcasts. While the results showed no overall difference by age groups, the 19-or-younger and 25-or-older age groups were more likely to listen to classroom podcasts than were the 20-24 age group.

Based on the results, it was concluded that students who listened to classroom podcasts were not motivated by age. Although the students listened to classroom podcasts for a variety of reasons, it was concluded that classroom podcasts were helpful for reviewing class lectures and hearing the interaction between teacher and students, regardless of the age of the students.

Conclusion and Finding for Research Question 5

Research question 5 asked, *Is there a significant difference between genders with regard to the number of hours per week students listened to classroom podcasts?*

The data showed that female and male students differed significantly in the measures used in this study. While the number of female and male students was roughly equal, more female students participated in classroom instruction supplemented with classroom podcasts and spent more time listening to classroom podcasts.

Based on the results, it was concluded that female students are more likely than male students to listen to classroom podcasts. The results suggest that female students have a more positive view of classroom podcasts. The ratio between female and male students could be regarded as a significant indicator for future consideration of technology-based teaching and learning.

Discussion

This study focuses primarily on the use of podcasting to distribute classroom materials to students via Ohlone's iTunesU, personal computers, and MP3 devices in Ohlone College. This pedagogical approach is a "listen and learn" learning concept. The goal is that students will adapt to today's mainstream culture, technological culture, and learning styles and be more engaged and motivated to learn. As described in Chapter 4, multiple factors were involved in this study. Five specific factors were examined: (a) pretest and posttest scores, (b) frequency of use of podcasts before and after the pretest, (c) gender, (d) class attendance, and (e) age. The pretest score, posttest score, frequency of use of podcasts, and gender were found to have significant effects on the frequency of use and attitudes about podcasting. Class attendance and age groups had no impact on frequency of use, but they are considered to be very beneficial to the findings and future research.

The first factor examines the differences in pretest and posttest scores. It was concluded that students who listened to classroom podcasts were more likely to receive higher test scores than those who did not. Based on the findings, it was concluded that use of classroom podcasts influenced test scores. However, the motivation of learners to receive higher test scores seems to be associated with self-directed learning and learners desired to take ownership of their own learning needs.

These kinds of learning are based on behaviorism, constructivism, and transformational learning theories, which are the theoretical framework for this study.

When students listened to classroom podcasts, they actively engaged in self-directed learning and reinforcement of what was being learned: behaviorism. Through the processes of adaptation, students were actively constructing new knowledge by listening to classroom podcasts: constructivist learning. When students make meaning of what are being learned through their experience with podcasting, they are utilizing a familiar tool (i.e., an entertainment device) to learn: transformational learning. This pedagogical approach utilizes today's mainstream culture, technological culture, and learning preferences to engage and reinforce student learning. Engaging in this kind of learning, students can utilize what they consider to be "fun and exciting" to learn subject matter.

The second factor examines the frequency of use of podcasts between pretest and posttest. The findings indicated that students who listened to classroom podcasts in preparation for the pretest were also more likely to listen to classroom podcasts in preparation for the posttest. Similarly, students who chose not to listen to classroom podcasts for the pretest were also more likely to listen to classroom podcasts for the posttest. There were not enough data to explain the reasons for this finding. The availability of classroom podcasts was likely to be one of the reasons. Based on the results, students who had lower test scores on the pretest were more likely to choose any available resource to receive higher test scores on the posttest. Again, students are more likely to be motivated by self-directed learning and the desire to take ownership of their learning needs.

Regardless of the countless dispositions, this study shows that students are likely to take advantage of any available resource to perform better on assessment

tests. Classroom podcasts reinforce and provide flexible learning outside the classroom. The results from this study suggest that (a) students who have lower midterm test scores are more likely to listen to classroom podcasts, (b) students who desire to receive higher course grades are inclined to listen to classroom podcasts, and (c) students who have positive attitudes toward podcasting and familiar with the technology are likely to favor classroom podcasts. Clearly, students who desire to achieve higher test scores are likely to use any available resources to achieve the desired outcome.

The third factor examines gender and frequency of use of podcasts. The findings showed that female students had more positive view of classroom podcasts. Traditionally, female students tend to have negative attitudes toward technology-based learning. At least in this study, female students were more likely than male students to use classroom podcast services. Giannini-Gachago and Seleka (2005, as cited in Moreau, 2009) reported that female students were more likely than male students to take part in discussions (as cited in Moreau, 2009, p. 448). Based on the open-ended questions in this study on the “quality of classroom podcasts and podcasting as learning tools,” most female students indicated that class lectures on podcasts were very helpful. This study showed significant gender differences in podcast use. According to Salkind and Rasmussen (2008), females were more achievement motivation and more likely to initiation action to meet their own learning needs. The majority of students in this study had positive views of classroom podcasts, especially the 19-or-younger age group and the 25-or-older age group.

The results show that podcasting could be a learning tool for students and teaching tool for teachers. However, podcasting is not for every student and not for every teacher familiar with the technology. As indicated by respondents to the handout

survey, not all students chose to participate in classroom instruction supplemented with classroom podcast. While most students took advantage of classroom podcasts, the Hispanic students had the lowest percentage rate of participation of all ethnic groups, while White and Asian students were more likely to take advantage of these resources available to them. The decision is clearly up to the individual student to decide whether to utilize podcast services.

The fourth factor examines class attendance and the frequency of use of podcasts. The findings showed that class attendance was not a factor in the frequency of use of podcasts. There were not enough data to explain this finding. However, based on responses to the open-ended question, the availability of classroom podcasts and the ease of download of the podcasts from Ohlone's iTunesU and on demand might be factors. Podcasting offers flexibility in learning outside the classroom; students can control *when* and *where* to listen and learn. This is what podcasting is about.

The fifth factor examines age groups and the frequency of podcast use. The findings indicated no significant difference in podcast use by age, although the 19-or-younger age group, mature students, and nontraditional students were more likely to engage in this form of learning. According to Salkind and Rasmussen (2008), adult learners are more likely to be “involved in things they like and less involved in pursuits they dislike” (p. 745). It was surprising that the 20-to-24 age group was less likely to listen to classroom podcasts, apparently preferring classroom lectures. From the open-ended question “learning preferences” category, a large number of respondents indicated that they preferred classroom lectures over classroom podcasts. Perhaps they were more confident and had more experience in knowing how to learn (p. 765).

As indicated in the five factors above, students who took advantage of the available resources to learn (not just classroom podcasts) are more likely to be more

engaged and motivated in the learning process and more likely to perform better on the assessment tests. This study indicates that classroom podcasts are helpful for student learning, especially for mature students and female students. Based on demographic profile data, low socioeconomic students and minority students had the lowest usage rates. Although there were no data to explain low participation rates by minority groups and social classes, it is reasonable to attribute these low rates lack of accessibility to technology. Based on the data generated from this study, part-time, nontraditional, and especially minority students were more likely to take advantage of the additional learning materials than were traditional college students.

Although this study did not directly examine podcasting as a teaching tool, it is considered to be an important factor. Based on course instructor feedback, there are positive and negative aspects to podcasting as a teaching and learning tool. Positive aspect include that (a) students can re-listen to the lectures as often as they like, (b) students can listen at their leisure, and (c) students who are struggling with learning note-taking skills and English as Second Language (ESL) students can benefit by podcast use. Disadvantages include that (a) students may rely on podcasting and not attend lectures, and (b) students download classroom podcasts but do not actually listen to them. As the course instructor stated, podcasting “is a great teaching tool because it will allow students access to class lectures whenever they want.”

A quasi-experimental design was an appropriate procedure for this comparison study of two intact groups; however, assignment of the treatment was not randomized, so the differences in results for the groups may have been due to differences in the groups either by subject area (History of the United States) or environmental conditions. This study lacked internal validity, which may have led to statistically significant results. Perhaps random sampling design is better to determine the true

nature of the differences. Participants in the study were anonymous and their test scores were also anonymous. However, random sampling design is not a practical procedure to study college students. Nonetheless, the quasi-experimental design represents naturally occurring differences between two groups (traditional classroom instruction and classroom instruction supplemented with classroom podcasts).

Recommendations

The main focus of this study was on the use of classroom podcasts to improve student learning. While the study was a case study of classroom podcast in Ohlone College, the data generated from the study could be used to provide leadership and support for future directions in technology-based instruction. As the Internet and Web-based technologies expanded learning opportunities for distance education, mobile and podcasting technologies have emerged as the new delivery of distance learning materials through the much more flexible learning medium. This study indicates that communication technology is at the fore of future distance education at all levels. Clearly, information technology has transformed every aspect of the information society (economic, education, financial, political, social). Therefore, to prepare for future education, the recommendation for school leaders is to invest in state-of-the-art technology infrastructure and faculty and staff training.

Podcasting is not possible without the technology infrastructure to support of the production, publication, and delivery of course content. Three key technology components support podcast services in Ohlone College: the Internet, Ohlone iTunesU, and the digital recording device. Having these technology components is the first step. Next, the decision to provide podcast services belongs to the course instructor. To provide audio/video podcast content for students, people in the leader-

ship positions have the power to encourage, support, and provide up-to-date teaching and learning methodologies, including technology-based training, and not just podcasting for their faculty and staff. To incorporate classroom learning materials through audio/video podcasts, the course instructor must be trained and familiar with the technology. Not all instructors are trained and equipped to provide podcast services. Based on training and an understanding of the benefits of podcasting, the course instructor will have enough information to decide whether podcasting is the appropriate instructional path.

The recommendation for the people in the leadership positions is to invest in faculty and staff educational training. For these and countless reasons, careful consideration of how best to incorporate technology-based learning, based on ongoing research, especially in the area of personal entertainment technologies, to determine the benefits and impacts of these technologies on student learning.

Many commercial and industrial businesses worldwide have recognized the importance of employee training and building organizational successes through well-trained and happy employees. While the goals of academic institutions are not the same as those for business, in today's information society the shift appears to be a convergence between businesses and academic institutions. Academic institutions can form partnerships with businesses to meet the needs of today's students (traditional students, nontraditional students, lifelong learners, continuous learners, etc.). Research has shown that effective teachers make a difference in student performance (Salkind & Rasmussen, 2008). That is, if the purpose of teaching is to improve student learning outcomes, it is prudent for educators to find the best available methods to inspire, engage, motivation, and reinforce students in the learning process. This is what Ohlone College's learning communities implied.

Podcast services cannot exist without necessary accessibility to students. The technology infrastructure must be on the cutting edge of technology to make these services available to faculty, staff, and students. The technology infrastructure is not limited to physical hardware/software components (Web servers, computer servers, hardware devices, operating software, etc.); the classroom learning environment must be equipped with Internet and other technology-based facilitate learning components.

There are many issues surrounding investment in state-of-the-art technology, such as facility, equipment, support, and maintenance. Human support staffs are important to make the technology available to stakeholders. The costs associated with having the right technology in place are extensive, and therefore further investigation is needed to determine the necessary balance between having state-of-the-art technology versus equipment and ongoing maintenance costs. Like Ohlone College, college and university leaders must determine their own needs for technology.

Suggestions for Future Research

Future research may replicate this study in other subject areas to test the overall effectiveness of classroom podcasts. One of the Ohlone College's (2004-2009) goals is to "develop across the curriculum the Learning College Model, utilizing methods and technologies that hold the most promise for improving student course and program completion success rates" (n.p.). This is a practical learning model for today's information society; therefore, it is recommended to promote and encourage faculty and staff to develop new ways to improve course and program completion rates. Podcasting classroom learning activities is one of these ways. This method can be replicated to study other subject areas to determine longitudinal effects and understand students' attitudes toward podcasting. Some of the students in the present

study expressed that it is a good idea to expand podcast services to other classes. As an alternative to Ohlone iTunesU, a new 3CRSS by 3C Media Solutions is less complex than the RSS format and can be used to provide audio/video podcast.

The next step of the student is to investigate the potential of video podcasting and personal entertainment devices for visual and interactive learners. As described in Chapter 2, new technologies continue to revolutionize the way people live, communicate, interact, work, and learn. Current trends in entertainment devices (cell phones, PDAs, games, and other digital devices) have opened unlimited possibilities for their use in higher education. When learning is “fun and exciting,” students are likely to be more engaged and motivated and to participate in the learning process, which will ultimately improve student learning outcomes. Some of the students in the study suggested that podcasting as a teaching and learning tool is very helpful for reviewing class lectures and hearing interaction between teacher and student. On the recommendation of the students, video podcasts is also needed for visual learners. Further study is needed to determine the impact of audio and video podcasts on visual learners.

An example of transforming consumer technology that has profoundly changed the mindset of how students learn, work, play, and interact is the Apple Computer (2010) multimedia tablet computer named the iPad, which has the ability to serve for Web surfing, e-reader, game device, audio/video player, and multipurpose multimedia device. Other leading manufacturers, such as Apple, Google, Motorola, and Microsoft, are striving to develop the next personal device. By adapting to today’s technological culture, educators can reach out to students.

The iPad and related smartphones are becoming more apparent in the everyday lives of today’s students. According to a survey conducted at Stanford University by

Professor Tanya Luhrmann in spring 2009 (as cited in Carey, 2009), the iPhone was very popular on campus, particularly among undergraduates students. An iStanford, the iPhone application was introduced in 2008 at Stanford University for helping students to navigate campus, with access provided to the campus directory, class schedules, email, map, search, and so forth. Since then, the application has been downloaded about 100,000 times by students, faculty, staff, visitors, and others. Nearly three quarters of respondents reported positive attitude toward their iPhones; 21% of the respondents stated that their iPhones (or smartphones) were more than just tech-toys and 17% described them as an extension of the brain. Keywords respondents used to describe their mobile phones included *fun* (66%), *cool* (61%), *addictive* (42%), *pretty* (38%), *classy* (29%), *magic* (25%), *urban* (25%), *appendage* (16%), *swank* (16%), *obsessed* (10%), *covet* (9%), *electronic leash* (8%), *drug* (7%), *standoffish* (5%), *fetish* (5%), and *obscene* (1%). The students' perceptions reflected by these keywords and their attitudes toward the iPhone were similar to those revealed in the current study.

Like the iPhone and MP3 player, the iPad will likely evolve into the next groundbreaking tool for student learning on the nation's college campuses. The Stanford survey emphasized the impact of the digital consumer technologies on student learning. Similarly, the current study showed that podcasting represents new thinking in the delivery of educational content for playback lectures and lessons on student mobile devices (or iPhones), MP3 players, or personal entertainment devices. The iPad seems to be the next major paradigm shift on the nation's college campuses. Further research about the effectiveness of the iPad is needed to explore how this medium can be used to facilitate learning, collaboration, and information sharing, as well as the impact of this device on student learning.

Conclusion

The findings of this study showed no difference between traditional classroom instruction and classroom instruction supplemented with classroom podcasts in regard to average midterm and final examination test scores. However, the course midterm and final examination scores of the group with classroom instruction supplemented with classroom podcasts were higher than those of the group with classroom instruction without classroom podcast, and the average final examination test scores were significantly higher than the average midterm test scores. As hypothesized, students who listened to classroom podcasts were more likely to achieve higher test scores than students who did not listen to classroom podcasts. The findings also showed that a large majority of the students, particularly female, part-time, nontraditional, and minority students, had very positive views of the podcasting service. Female students were significantly more likely to use the podcasting service, which is a finding worthy of further investigation. In terms of podcast listening devices, the findings showed that the large majority of respondents owned a PC, MP3, or related PDA device.

The information from this study can be helpful for school leaders and educators to determine how podcasting and related personal entertainment technologies could be used to engage, motivate, and reinforce student learning. As indicated by this study, podcasting has the potential to improve student learning but it is not for every student. Furthermore, the information from this study can be very helpful for school leaders and educators in understanding contemporary society and the importance of technology integration in educational institutions and the teaching and learning environment.

Podcasting has made a contribution to the field of education by creating a learning condition for students who need extra help, especially students with physical,

social, emotional, and intellectual needs. As these students use the thinking tool in the form of personal entertainment and leisure, they are engaged and learning more effectively. Students' learning styles are important to their learning outcomes. By leveling the playing field for students and their learning preferences, these devices may unleash learning potential and learning effectiveness. Podcasting pedagogy is particularly promising for faculty to digitally record and distribute lectures and other information to students beyond classroom instruction. Overall, this approach may be considered as the next step forward in technology-based instruction. In this context, the pedagogy approach to teaching and learning is essentially taking what was already there into another level.

One student in the study stated that podcasting is "a huge technological step in the right direction for students." If podcasting or any newly emerged technology has a potential to be used to engage, empower, inspire, and motivate student learning, then educators have the obligation to explore the possibilities. This is essentially what the National Education Association stated in its Principle I, Commitment to the Student: "The educator strives to help each student realize his or her potential as a worthy and effective member of society" (as cited in Rebores, 2001, p. 171). By exploring new and better ways of teaching and learning, students will be more prepared to become productive members of society and to enter the new age of the information society. The Student Services Division at Ohlone College (2010), the site for this study, had a similar charter statement of commitment to the student: "Student Services as a division exists to focus on the whole student, and the entire student learning experience."

With the phenomenon of mobile communications and personal entertainment technologies in academia, it seems that traditional form of instruction should take the

blended approach to instruction. By reaching out in today's mainstream culture, technological culture, and learning styles of students, educators can understand how to help students to realize their full learning potential. This study has shown that traditional classroom instruction does not change but has evolved into a range of technology-assisted learning methods for student learning, far beyond podcasting. This evolution of teaching and learning pedagogy was described by Robins and Webster (2002) as "the emerging and applied user-shaped kind of knowledge that is now required" in the modern information society (p. 42).

As they move education into the 21st century and the new age of the information society, educators should consider technology-based teaching and learning as the leader in student learning. Recurring training programs for faculty and staff are essential to conduct technology-based teaching and learning effectively. School leaders have the obligation to provide educators the necessary tools and equipment to facilitate learning activities. Clearly, as shown in this study, greater innovations of information technology, not only podcasting, have altered the delivery of education, training, and professional development. With increasing domination by information and communication technologies, academia is trying to redefine its image beyond just the site for teaching and learning. The integration of technology into education in the 21st century seems to be a natural order in the modern information society; however, careful consideration by educators is necessary to determine the right balance between technology and the sole reason for teaching and learning.

The World Summit on the Information Society (WSIS, 2003) stated that it is a global responsibility to "create access, utilize and share information and knowledge, enabling individuals, communities and peoples to achieve their full potential" (n.p.). A bold response to the last question in the survey by another student was that "classroom

podcasts should be made available to the world.” Such a position is reasonably to be expected from the world’s leading organization on the information society; however, even a community college student has understood the meaning of the world of information sharing.

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APPENDICES

APPENDIX A
DATA DICTIONARY

Table A1

Data Dictionary for Demographics Variables Descriptions

Variable	Type	Assigned value	Scale
Gender	Dichotomous	1 = Male 2 = Female 3 = Other (self-describe)	Nominal
Age group	Categorical	1 = 19 years and younger 2 = 20 to 24 years 3 = 25 to 39 years 4 = 30 to 34 years 5 = 35 to 39 years 6 = 40 to 49 years 7 = 50 years and older	Ordinal
Ethnicity	Categorical	1 = African American 2 = Asian 3 = Filipino 4 = Hispanic 5 = Pacific Islander 6 = White 7 = Other	Nominal
Year in school	Numeric	1 = First year 2 = Second year 3 = Third year 4 = Fourth year 5 = Fifth year or higher	Ordinal
Academic major	Categorical	1 = Arts and sciences 2 = Business 3 = Engineering 4 = Other	Nominal
Class attendance	Categorical	1 = less than other classes 2 = more than other classes 3 = same as other classes	Nominal

Table A2

Data Dictionary for Podcasting-Related Variables Descriptions

Variable	Type	Assigned value	Scale of measurement
Listening device	Categorical	1 = MP3 player 2 = Cell phone/PDA 3 = PC 4 = PC and MP3 player	Nominal
Perceived ease of use	Categorical	1 = Strongly disagree 2 = Disagree 3 = Agree 4 = Somewhat agree 5 = Strongly agree	Ordinal
Perceived usefulness	Categorical	1 = Strongly disagree 2 = Disagree 3 = Agree 4 = Somewhat agree 5 = Strongly agree	Ordinal
Motivation after the midterm	Categorical	1 = Strongly disagree 2 = Disagree 3 = Agree 4 = Somewhat agree 5 = Strongly agree	Ordinal
The number times listen to before the midterm	Numeric	Number value	Scale
The number of times listen to after the midterm	Numeric	Number value	Scale
The number of times listen to the same podcast	Numeric	Number value	Scale
The frequency of use	Numeric	Number value	Scale

Table A3

Data Dictionary for Midterm and Final Examination Test Scores Variables Descriptions

Variable	Type	Assigned value	Scale of measurement
Midterm test scores	Numeric	Number value	Scale
Final examination test scores	Numeric	Number value	Scale
Final course grade	Numeric	5 = A 4 = B 3 = C 2 = D 1 = F	Scale

APPENDIX B
OHLONE ITUNESU

Ohlone iTunesU - Ohlon... x

http://itunes.ohlone.edu/

Education Personal News Misc Yahoo! Windows Live Other bookmarks

OHLONE COLLEGE

Ohlone iTunesU

- Download iTunes
- Students**
 - Launch iTunesU
- Faculty**
 - iTunesU Faculty Login**
 - How to upload files - Tutorial

A-Z Index | Search | Directory | Ohlone Home

Ohlone iTunesU

Welcome to Ohlone College's iTunesU.

Ohlone's iTunesU allows you to access and download audio and video podcasts to your computer or iPod directly from iTunes.

Fall 2006 began our pilot rollout of iTunesU.

We invite you to experience this exciting learning tool by clicking on the link below.

[Take me to Ohlone on iTunesU](#)
(this will launch iTunes)



iTunesU requires the latest version of iTunes from Apple to be installed.
[Download iTunes from Apple.](#)

[iTunesU Faculty Login](#)

APPENDIX C

HIST-117A CLASSROOM PODCASTS

File Edit View Controls Store Advanced Help

iTunes

Search Ohlone College

LIBRARY

- Music
- Movies
- TV Shows
- Podcasts
- Radio

STORE

- iTunes Store
- SHARED
 - Home Sharing
- GENIUS
- PLAYLISTS
 - iTunes DJ
 - 90's Music
 - Classical Music
 - Music Videos
 - My Top Rated
 - Recently Added
 - Recently Played
 - Top 25 Most Played
 - WALT DISNEY'S GR...

Ohlone College > History > U.S. History 117A - Prof. Darren L. Bardell - Tab 1

U.S. History 117A - Prof. Darren L. Bardell

LOGOUT

Last Modified: Dec 10, 2009
Tracks in Tab 1: 19

GET TRACKS
SUBSCRIBE

Tab 1 Tab 2

	Name	Time	Artist	Album	Price
1	The New England Way	1:17:08	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
2	Mercantilism & Enlightenment	53:15	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
3	Ben Franklin & Stage 1 of Rev	1:19:10	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
4	Ch 6/2 part 1: Road To Revolution 2/26	50:06	h117 General Podcast	U.S. History 117A - Prof. Darren L. Bardell	Free GET
5	Ch 6/2 part 2: Phases of Revolution 2/28	52:11	h117 General Podcast	U.S. History 117A - Prof. Darren L. Bardell	Free GET
6	Final Phases to Revolution	1:20:22	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
7	Revolutionary War	1:13:31	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
8	The Experiment Begins	1:20:29	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
9	President Washington	1:31:22	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
10	Jefferson's Revolutions	39:36	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
11	Early National Period	1:13:12	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
12	Mainfest Destiny	1:04:22	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
13	American Exceptionalism	1:25:58	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
14	Economic Development North v South	1:24:42	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
15	War With Mexico	1:32:46	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
16	Prelude to Civil War	1:26:08	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
17	Civil War part 1	1:02:40	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
18	Civil War Part 2 + Reconstruction	1:27:42	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET
19	Reconstruction And Final exam Review	1:15:27	Ohlone College	U.S. History 117A - Prof. Darren L. Bardell	Free GET

19 songs

APPENDIX D
DATA COLLECTION INSTRUMENT

Data Collection Instrument

The questions below will be used to assist researchers in data collection regarding the use of classroom podcasting in Ohlone College. By completing this survey, you will enable researchers to study how podcasting can be used to improve teaching and learning at Ohlone College. The questionnaire was organized into three parts: (a) demographics, (b) podcasting, and (c) open-ended questions.

Demographic Questions

1. What is your gender? (*Check one*)
☐ Male ☐ Female ☐ Other (self-describe)
2. What is your age group? (*Check one*)
☐ 19 years and younger ☐ 20 to 24 years ☐ 25 to 29 years ☐ 30 to 34 years
☐ 35 to 39 years ☐ 40 to 49 years ☐ 50 years and older
3. What is your major/academic? (*Check one*)
☐ Arts ☐ Sciences ☐ Business ☐ Engineering
☐ Other, please specify: _____
4. What is your year in college? (*Check one*)
☐ First year ☐ Second year ☐ Third year ☐ Fourth year ☐ Fifth year or higher
5. What is your ethnic background? (*Check one*)
☐ African American ☐ Hispanic or Latino
☐ Asian ☐ Native American
☐ Filipino ☐ White Non-Hispanic
☐ Multi-racial ☐ Other, please specify: _____

Podcasting Questions

6. Prior to this course, were you familiar with a term *podcasting* and what it is?
(Check one)

☐ Yes ☐ No

7. Which of the following listening devices did you use to listen to the podcasts?
(Check all that apply)

☐ MP3 player or Apple iPod ☐ Cell phone, PDA or other handheld device
☐ Personal Computer or Laptop ☐ I don't have any of these devices

8. Was it easy to access, download, and replay audio podcasts from instructor's Website or Ohlone iTunesU? (Circle the number that reflects your opinion)

1	2	3	4	5
Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree

9. Listening to audio podcasts helped my understanding of the course subject matter.
(Circle the number that reflects your opinion)

1	2	3	4	5
Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree

10. After the midterm, were you more motivated to listen to classroom podcasts because you want to improve your final exam test scores? (Circle the number that reflects your opinion)

1	2	3	4	5
Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree

11. After the midterm, how many times did you listen to classroom podcasts for this course?
(Check one)

☐ One ☐ Twice ☐ Three ☐ Four or more times ☐ Not at all listen to podcast

12. Before the midterm, how many times did you listen to classroom podcasts for this course? (Check one)

☐ One ☐ Twice ☐ Three ☐ Four or more times ☐ Not at all listen to podcast

13. How many times did you listen to the same podcasts for this course? (Check one)

☐ One ☐ Twice ☐ Three ☐ Four or more times ☐ Not at all listen to podcast

14. In an average week, how much time did you spend listening to podcasts for this course? (*Check one*)

☐ 1 hour or less ☐ 2 hours ☐ 3 hours or more ☐ Not at all listen to podcast

15. What best described your class attendance compare with your other classes? (*Check one*)

☐ I attended class LESS than other classes

☐ I attended class MORE than other classes

☐ About the same attendance as other classes

Open-Ended Questions

16. How could classroom podcasts be improved to better meet your needs?

APPENDIX E
PERMISSION TO CONDUCT THE STUDY GRANTED BY
THE
OFFICE OF STUDENT SERVICES, OHLONE COLLEGE



A World of Cultures United in Learning

Ohlone Community College District

43600 Mission Boulevard
Fremont, CA 94539-0390

Ohlone College Newark

Center for Health Sciences & Technology
39399 Cherry Street, Newark, CA 94560

www.ohlone.edu

May 27, 2009

Institutional Review Board
Alliant University
One Beach Street, Suite 100
San Francisco, CA 94133

To Whom It May Concern:

Please note that Mr. James Pham, AIU graduate student, has the permission of Ohlone College to conduct research at our campus for his study on perceived effects of podcast use in the classroom. Mr. Pham has my permission, as the designated official for student (human) subject research, to contact students via email in five course sections.

If there are any questions, please contact my office at (510) 659-6262.

Sincerely,

Ron Travenick
Vice President, Student Development

APPENDIX F
INSTRUCTORS' CONSENT TO PARTICIPATE



OHLONE
COLLEGE

A World of Cultures United in Learning

Ohlone Community College District
43600 Mission Boulevard
Fremont, CA 94539-0390

Ohlone College Newark
Center for Health Science &
39399 Cherry Street Newark, CA 94560

www.ohlone.edu

August 27, 2009

Institutional Review Board
Alliant International University
One Beach Street, Suite 100
San Francisco, CA 94133

To Whom It May Concern:

Mr. Jim D. Pham, AIU graduate student who is a Computer Science instructor at Ohlone College, has my permission to use my class, HIST-117A, History of the United States for his study on the use of classroom podcasts as supplemental course materials for playback lectures and other learning lessons and the relationship its have with the final course grade for students in Ohlone College.

If there are any questions, please contact my office at (510) 316-0726.

Sincerely,

Darren Bardell
Associate Professor
Department of History, Ohlone College
43600 Mission Blvd.
Fremont, CA 94539
Home Page: <http://www.ohlone.edu/people/dbardell/>

APPENDIX G
STUDENTS' CONSENT TO PARTICIPATE

Informed Consent Letter for Students

Title of study: A Case Study of Classroom Podcast in Ohlone College

Researcher: Jim D. Pham

I am a graduate student working on a case study of classroom podcast in Ohlone College for my class project at Hufstедler School of Education, Alliant International University, in San Francisco, California. With your instructor's permission, during the Fall 2009 semester, History of the United States (HIST-117A) is selected because of its large class size of 120 to be used for the study.

Through data collection, I will study the differences between traditional educational instruction and classroom instruction supplemented with classroom podcasts. The questionnaire you will be completing is organized into three parts: (a) demographics, (b) podcasting, and (c) open-ended questions. It will take approximately 15 minutes to complete. The information will be used to assist researchers like myself to study the effectiveness of the use of classroom podcasting. I will be writing a thesis paper describing my findings in partial fulfillment of the requirements for the degree of Doctor of Education.

You will be an anonymous participant. Your individual responses to the survey/questionnaire at the end of the course will be kept confidential and will be destroyed after the completion of the study. Your name will not be used in any written documents associated with this study. The data collected and the paper written may help future educators to determine the necessary balance between podcast learning and traditional classroom interaction. You are therefore encouraged to represent your true feedback as honestly as possible.

Participation in this study is voluntary. You may choose not to participate in the handout survey/questionnaire at the end of the course. Your instructor will not know who did or did not participate in the study, and there is NO relationship between grade you receive in this course, credit you receive in this course, and whether you do or do not participate. You may withdraw at any time, no questions asked, you do not have to complete the questionnaire. You are subject to minimal risk as a participant in this study since your name will not be used in association with any documents.

I appreciate your willingness to participate in this project. Let me know if you have any questions about your participation, now or in the future. You may also contact my advisor, George Stamos, Ed.D by email gstamos@alliant.edu, or the IRB (Institutional Review Board) at Alliant International University in San Francisco at 415.955.2151. You may also contact your instructor for the result of this study by email dbardell@ohlone.edu. The result of this study will be available around October, 2010. Please know that you are making a positive contribution to my education.

By submitting your handout survey/questionnaire, you have agreed to participate in this study. I will provide you with a copy of this form for your records.

Sincerely,



Jim D. Pham
Computer Science Instructor, Ohlone College
43600 Mission Blvd., Fremont, CA 94539 (408) 623-5211

APPENDIX H
RISK ASSESSMENT BY THE DISSERTATION COMMITTEE

RISK ASSESSMENT LEVELS BY THE DISSERTATION COMMITTEE

Jim D. Pham	448224	jpham@ohlone.edu
Student Name (typed)	Student ID#:	E-mail address
Student Telephone Numbers:	408-528-8958	408-623-5211
	home	Office

Title of Proposed Research: A Case Study of Classroom Podcast in Ohlone College

Use the following scale to assess risk to human subject(s) of proposed research procedures

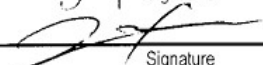
1 = NO RISK* 2 = MINIMAL RISK** 3 = MODERATE RISK*** 4 = HIGH RISK***

* No risk studies are those which have no social, psychological, or physical danger to subjects.

** Minimal risk implies risk no greater than that of typical daily experience and includes taking standardized psychological tests or routine forms of examination and treatment.

*** Levels 3 & 4 must be reviewed by the IRB Committee (currently meets the first Friday of each month, except summer months).

RISK LEVELS MUST BE ASSIGNED AS AN INTEGER (NOT AS 1.5, 2.5, ETC.)

Jim D. Pham		2
Printed Name of Student	Signature	Risk Assessment Scale#
George Stamos, EdD		2
Printed Name of Chairperson	Signature	Risk Assessment Scale#
Joseph Adwere-Boamad, PhD		2
Printed Name of Reader #1	Signature	Risk Assessment Scale#
James Hiramoto, PhD		2
Printed Name of Reader #2	Signature	Risk Assessment Scale#